SE 3002 — Software Quality Engineering

FAST School of Computing — Department of Software Engineering — FAST NUCES Islamabad

ASSIGNMENT 3 — Student Submission Template

Black-Box Testing: EP & BVA on Testfolio

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Section / Batch	**SE C / Batch 2024**
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Test Account Used	**i243137@isb.nu.edu.pk**
Browser & Version	**Google Chrome 156.0.8078.47 on Windows 11 (64-bit)**

ℹ *This template is pre-formatted. Add rows to any table as needed. Do NOT delete section headings or column headers. Export to PDF before submitting.*

⚠ **Italic gray text = fill-in prompt. Replace every prompt with your own content before submitting.**

D1 — Test Plan [20 marks]

TP-1 Test Plan Metadata [1 marks]

Test Plan ID	**TP-A3-24i-3137**
Version	**1.0 (Final)**
Author (Roll No.)	**Umair Hassan (24i-3137)**
Status	**Complete / Final**

TP-2 Introduction [2 marks]

Briefly describe Testfolio and the purpose of this test plan:

Testfolio (https://testfol.io/) is an advanced, web-based financial backtesting, asset allocation, and scenario modeling platform. It empowers quantitative researchers and individual investors to evaluate investment strategies across historical datasets, simulate retirement cashflow withdrawals, compute efficient frontiers, and optimize multi-asset portfolios.

The purpose of this software quality engineering test plan is to conduct rigorous black-box functional testing of Testfolio's deployed web application. By applying Equivalence Partitioning (EP) and Boundary Value Analysis (BVA), this testing effort systematically verifies client-side and server-side validation rules, identifies boundary failure modes across continuous and discrete domains, evaluates system robustness under invalid inputs, and provides an authentic, evidence-backed test record to guarantee compliance with SE 3002 standards.

TP-3 Scope [2 marks]

In-Scope Features (list every form and input field you will test):

The testing scope encompasses all eleven (11) interactive tools and forms deployed on Testfolio as mandated by the assignment handout:
1. Create Account (Sign Up) — https://testfol.io/start (modal: email address, password, client validation, auth provider)
2. Sign In (Login) — https://testfol.io/login (modal: email address, password, authentication feedback, one-time code)
3. Portfolio Backtester (CRITICAL FORM) — https://testfol.io/ (parameters: date range, start/end dates, starting value, rolling window, cashflow legs, asset tickers, allocations %, rebalance frequencies)
4. Monte Carlo Simulator — https://testfol.io/monte-carlo (simulation years/horizon, iterations/count, asset distributions)
5. Safe Withdrawal Rate Calculator — https://testfol.io/safe-withdrawal-rate-calculator (initial capital, withdrawal rate %, duration, asset weights)
6. Backtest Optimizer — https://testfol.io/backtest-optimizer (target assets, min/max asset weight constraints, optimization metrics)
7. Efficient Frontier — https://testfol.io/efficient-frontier (multi-asset covariance matrix, risk-return bounds, asset count >= 2)
8. Contact Form — https://testfol.io/contact (name, email address, topic category, subject line with counter 0-120, message body with counter 10-5000, Cloudflare Turnstile anti-abuse check)
9. Lump Sum vs DCA — https://testfol.io/lumpsum-vs-dca (initial lump sum $, periodic DCA contribution $, investment interval, asset selection)
10. Time Value of Money Calculator — https://testfol.io/time-value-of-money (contribution frequency, contribution timing, X/Y/Color variables, starting value $, monthly contribution $ between $0 and $100,000)
11. Rebalancing Sensitivity — https://testfol.io/rebalancing-sensitivity (rebalancing schedules: Monthly/Quarterly/Yearly/Never, rebalance bands, asset drift).

TP-4 Test Objectives [2 marks]

The primary technical objectives of this test engineering effort are:
1. Verify Functional Correctness: Ensure that valid financial parameters, date ranges, asset allocations, and credentials produce accurate mathematical calculations and simulations without runtime errors.
2. Evaluate Robustness & Error Trapping: Validate that invalid inputs (negative values, out-of-range percentages, inverted date ranges, malformed emails, short messages) are trapped gracefully with clear error banners or tooltips, preventing invalid calculation execution.
3. Determine Boundary Exactness: Empirically identify and probe exact mathematical and syntactic thresholds (e.g., minimum initial investment = $1, allocation sum = exactly 100%, Start Date < End Date, TVM monthly contribution = $0 to $100,000, contact message length = 10 to 5,000 characters).
4. Assess Security & Authentication Controls: Verify client-side prevention of malformed requests and server-side responses for unauthorized access or unauthenticated usage on gated features.
5. Maintain 100% Traceability and Evidence: Produce an auditable execution log supported by authentic, live screenshots bearing the student roll number sticky note to demonstrate genuine verification.

TP-5 Test Approach [3 marks]

Describe how you applied EP and BVA, and your probe strategy:

1. Equivalence Partitioning (EP) Approach:
Every in-scope input field across all 11 forms was partitioned into mutually exclusive, exhaustive subsets of input data. Partitions were classified into Valid Equivalence Classes (VEC) representing expected operational states and Invalid Equivalence Classes (IEC) representing error conditions, boundary breaches, or malformed data. Unique sequential identifiers (EC-001 through EC-074) were assigned. Weak EP with Robustness was applied across all forms to guarantee that every valid and invalid class is exercised by at least one dedicated test case. Strong EP was applied to the designated Critical Form to evaluate complex combinatorial interactions.

2. Boundary Value Analysis (BVA) Approach:
BVA was applied to all fields with ordered numeric, date, percentage, or length domains. Each boundary was assigned a unique sequential ID (BC-001 through BC-015). For all standard forms, 2-point BVA was conducted (testing values at the boundary and immediately adjacent). For the designated Critical Form, 3-point BVA was applied, rigorously testing Below, At, and Above positions for every identified boundary.
Fields with nominal enumerations (e.g. rebalancing frequency dropdowns), binary discrete toggles (e.g. inflation checkboxes), and cryptographic verification states (Cloudflare Turnstile) were excluded from BVA with documented engineering justification.

3. Designation of Critical Form (Portfolio Backtester — testfol.io/):
The Portfolio Backtester is designated as the Critical Form due to its architectural role as the core functional engine of Testfolio and its multi-variable combinatorial constraints. It requires simultaneous validation of asset allocation percentages summing to exactly 100%, relational date constraints (Start Date < End Date), positive starting capital (>= $1), and multi-asset ticker configurations. This complexity justifies both Strong EP and 3-point BVA.

4. Exploratory Probe Testing Strategy:
Because Testfolio is a live, proprietary web application with intentionally withheld specification documents, exploratory probe testing was conducted prior to formal test case authoring. A structured sequence of 16 empirical probes was executed across forms to discover client-side HTML5 validation rules, server-side toast banners, DOM state modifications, and hidden boundary limits.

TP-6 Test Environment [2 marks]

Browser & Version	**Google Chrome 156.0.8078.47 (Official Build, 64-bit)**

Operating System	**Microsoft Windows 11 Home / Pro (Build 22631, 64-bit)**
Screen Resolution	**1280 × 720 (Primary Display Resolution)**
Test Account (email)	**i243137@isb.nu.edu.pk (Authentic FAST Student Account)**
Network / VPN	**Direct Broadband / University Network; No Proxy; No VPN**

TP-7 Entry & Exit Criteria [2 marks]

Entry Criteria (conditions that must be true before testing begins):

Testing shall commence only when all of the following Entry Criteria are satisfied:
1. Testfolio web application is deployed, live, and publicly accessible at https://testfol.io without network outage.
2. Student testing account (i243137@isb.nu.edu.pk) is registered and operational for gated features.
3. Primary browser (Google Chrome 156 on Windows 11) is configured at 100% zoom with DevTools accessible.
4. Official assignment handout and template documents are loaded and structured in workspace.
5. Digital roll number sticky note (24i-3137) is configured and verified visible for screen capture.

Exit Criteria (conditions that mark testing as complete):

Testing shall be deemed complete only when all of the following Exit Criteria are satisfied:
1. All 11 in-scope forms and tools are fully investigated and inventoried.
2. Minimum 15 exploratory probes are executed, analyzed, and documented in TP-10 Probe Log (16 probes achieved).
3. All input fields across all 11 forms are partitioned into unique sequential EC IDs with VEC/IEC classifications in Layer 1.
4. Combined EP Test Case Table includes Weak EP with Robustness for all forms plus Strong EP for the critical form (77 EP test cases).
5. BVA Boundary Tables (Layer 1) define all applicable continuous boundaries with unique sequential BC IDs (15 boundaries).
6. Combined BVA Test Case Table covers 2-point BVA across forms and 3-point BVA (Below, At, Above) on the critical form (32 BVA test cases).
7. Traceability Matrix achieves 100% mapping across all 109 test cases with execution statuses and verdicts.
8. Authentic screenshots bearing the student roll number sticky note are embedded for every executed test case in the Execution Log.
9. Final document is formatted in DOCX and compiled to official PDF named A3_24i-3137_UmairHassan.pdf.

TP-8 Test Schedule [2 marks]

Activity	Notes / Dates	Est. Hours	Done?
Probe / Explore all 11 Testfolio forms	Oct 24–25, 2026; live exploratory probing & screenshots	4.0	**Yes**
Complete EP Partition Tables (Layer 1)	Oct 25, 2026; defined EC-001 to EC-074 across all fields	3.0	**Yes**
Design EP Test Cases (Layer 2)	Oct 25, 2026; Weak EP + Strong EP for Backtester (77 TCs)	3.5	**Yes**
Complete BVA Boundary Tables (Layer 1)	Oct 25, 2026; defined BC-001 to BC-015 with justifications	2.5	**Yes**
Design BVA Test Cases (Layer 2)	Oct 25, 2026; 2-point BVA + 3-point BVA on Backtester (32 TCs)	3.0	**Yes**
Execute test cases & capture screenshots	Oct 25–26, 2026; live execution on Chrome with roll sticky note	5.0	**Yes**

Activity	Notes / Dates	Est. Hours	Done?
Complete traceability matrix	Oct 26, 2026; mapped 109 test cases to fields, techniques, verdicts	2.0	**Yes**
Compile PDF and submit	Oct 26, 2026; document formatting, PDF export, LMS ready	1.5	**Yes**

TP-9 Risks & Mitigations [2 marks]

Risk	Likelihood / Impact	Mitigation
Live Website Dynamic Updates: Testfolio is an actively updated web app; UI changes or ticker data updates could alter input validation rules mid-testing.	**Medium / High**	Lock test environment to specific probe sessions; record timestamped network requests; verify all boundaries using consistent historical tickers (VTISIM, BNDSIM).
Third-Party Rate Limiting & Anti-Abuse Blocking: Repeated contact form submissions or backtest runs could trigger Cloudflare rate limiting or captcha lockouts.	**Low / High**	Space automated/manual requests with realistic human pauses; use valid educational credentials; solve Turnstile challenges legitimately; test edge values sequentially.
Client-Side vs Server-Side Validation Discrepancies: HTML5 native tooltips may mask underlying server responses or silently handle unexpected input characters.	**Medium / Medium**	Examine both client DOM interaction (button disabled states, form constraints) and server toast banners; test bypassed inputs; document distinct failure tiers explicitly.

TP-10 Probe Log — Appendix A [2 marks]

ℹ *Record each probe you performed while exploring the application. For each row, list ALL inputs entered — include context fields (fields needed to submit the form) alongside the field under investigation. Minimum 15 probes.*

Probe ID	Form	Test Inputs (field: value, …)	System Response	Inference / Note
PR-001	Sign Up Modal (/start)	Target: Password: '' (empty) Context: Email: 'probe_check@test.com'	Sign in button is completely disabled (dimmed blue); form submission blocked without error banner.	Password is mandatory prerequisite; client disables button rather than showing banner. (Evidence: TP10-P01)
PR-002	Sign In Modal (/start)	Target: Password: 'a' (1 char) Context: Email: 'test@example.com'	Sign in button enables at 1 char. Submitting triggers red banner: 'Email or password wasn't recognized. Try again...'	Min length boundary to activate submit is 1 character. Backend returns secure generic authentication error. (Evidence: TP10-P02)
PR-003	Sign In Modal (/start)	Target: Email: 'notanemail' (no @) Context: Password: dummy	Clicking Sign in triggers HTML5 tooltip: 'Please include an '@' in the email address. 'notanemail' is missing an '@'.'	Email field enforces native type='email' validation; traps syntax errors client-side before network dispatch. (Evidence: TP10-P03)
PR-004	Portfolio Backtester (/)	Target: Portfolios: 0 (empty list) Context: Default parameters	Clicking BACKTEST immediately triggers red banner: 'No valid	System enforces mandatory portfolio prerequisite; backtesting engine requires >=

Probe ID	Form	Test Inputs (field: value, …)	System Response	Inference / Note
			portfolios were found.' Execution halted.	1 configured portfolio. (Evidence: TP10-P04)
PR-005	Portfolio Backtester (/)	Target: Allocations sum = 120% (VTISIM: 60%, BNDSIM: 60%) Context: Default 60/40 preset	Total allocation turns red (120%). Clicking BACKTEST triggers red banner: 'Portfolio 1: Allocation percentages must sum to 100'.	Strict equality boundary: Total portfolio asset allocation must equal exactly 100%. (Evidence: TP10-P05)
PR-006	Portfolio Backtester (/)	Target: Starting Value: -$7 Context: Valid 60/40 portfolio	Clicking BACKTEST triggers validation tooltip: 'Value must be greater than or equal to 1.' Backtest blocked.	Lower bound identified: initial capital must be >= 1. Negative amounts trapped by client range constraint. (Evidence: TP10-P06)
PR-007	Portfolio Backtester (/)	Target: Starting Value: $0 Context: Valid 60/40 portfolio	Clicking BACKTEST triggers validation tooltip: 'Value must be greater than or equal to 1.' Backtest blocked.	Conclusive BVA evidence: 0 is 'Below' boundary and invalid; exact boundary value is 1. (Evidence: TP10-P07)
PR-008	Portfolio Backtester (/)	Target: Starting Value: '' (empty) Context: Valid 60/40 portfolio	Clearing field and clicking BACKTEST displays top red banner: 'Invalid starting value'.	Empty input is a distinct invalid equivalence class handled via application banner. (Evidence: TP10-P08)
PR-009	Portfolio Backtester (/)	Target: Start: 01/01/2023, End: 01/01/2020 Context: Custom range, 60/40 portfolio	Clicking BACKTEST displays red banner: 'Start date must be before end date'. Execution halted.	Chronological order strictly enforced; End date preceding Start date is rejected. Relational bound: Start < End. (Evidence: TP10-P09)
PR-010	Time Value of Money (/time-value-of-money)	Target: Form layout exploration Context: Live page load	Discovered real fields: Contribution Frequency/Timing, Variables (X, Y, Color), Range toggles, Starting Value $, Monthly contribution $.	Verified interface structure; non-standard TVM parameterization. (Evidence: TP10-P10)
PR-011	Time Value of Money (/time-value-of-money)	Target: Unauthenticated access Context: Click CALCULATE without login	Modal dialog opens: 'Sign in required. Sign in with a free account to use the Time Value of Money Calculator.'	Advanced analytical tools enforce free account authentication gating. (Evidence: TP10-P11)
PR-012	Time Value of Money (/time-value-of-money)	Target: Monthly contribution: -$500 Context: Signed in, $10,000 starting value	Clicking CALCULATE displays red banner: 'Contribution must be between $0 and $100,000.' Execution blocked.	Explicit numeric boundary discovered: Lower bound = $0, Upper bound = $100,000. (Evidence: TP10-P12)
PR-013	Time Value of Money (/time-value-of-money)	Target: Monthly contribution: $0 Context: Default parameters	Calculation succeeds! Renders interactive curve: 'Results (sampled 1805 feasible solutions)'.	BVA lower bound confirmation: $0 is 'At' boundary and VALID. Range is closed [0, 100000]. (Evidence: TP10-P13)
PR-014	Time Value of Money (/time-value-of-money)	Target: Monthly contribution: '' (empty) Context: Following PR-013 execution	Displays banner: 'Contribution must be between $0 and $100,000.' Prior chart results remain rendered on screen.	Usability defect: Prior valid graph is not cleared when subsequent input validation fails. (Evidence: TP10-P14)
PR-015	Contact Form (/contact)	Target: Message: 'ijklmno' (7 chars) Context: Email: i243137@isb.nu.edu.pk, Subject: 'abcdefg'	Displays red banner: 'Message must be at least 10 characters.' Counter displays 7 / 5000.	BVA lower bound identified: min message length = 10 chars. Lengths < 10 are rejected. Max is 5,000. (Evidence: TP10-P15)
PR-016	Contact Form (/contact)	Target: Message: >= 10 chars Context: Email:	Green banner displays: 'Message sent. We will get back to you soon.' Form resets.	Valid payload accepted; Turnstile anti-abuse token verified by server. (Evidence: TP10-P16)

Probe ID	Form	Test Inputs (field: value, …)	System Response	Inference / Note
		i243137@isb.nu.edu.pk, Captcha passed		

D2 — Equivalence Partitioning [25 marks]

ℹ *D2 has two layers. Complete Layer 1 first — one partition table per input field. Then write Layer 2 — your EP test cases, citing the EC IDs from Layer 1.*

LAYER 1 — EP Partition Tables (one table per input field — identify equivalence classes and assign EC IDs)

ℹ *One partition table per input field across all 11 forms. Assign EC IDs sequentially (EC-001, EC-002, …) across all fields. Add more tables as needed.*

EP-01 Feature / Form: Auth Modal (/start) Input Field: Email address

EC ID	Equivalence Class Description	Representative Value	Type (VEC / IEC)
EC-001	Valid standard email format (contains @ and domain)	test@example.com	**VEC**
EC-002	Registered student FAST university email format	i243137@isb.nu.edu.pk	**VEC**
EC-003	Malformed email string missing '@' symbol	notanemail	**IEC**
EC-004	Malformed email missing top-level domain	user@domain	**IEC**
EC-005	Completely empty / blank email address	(empty)	**IEC**

EP-02 Feature / Form: Auth Modal (/start) Input Field: Password

EC ID	Equivalence Class Description	Representative Value	Type (VEC / IEC)
EC-006	Valid alphanumeric password (length >= 8)	Testfol123!	**VEC**
EC-007	Single-character input (activates submit button)	a	**VEC**
EC-008	Password with special characters and symbols	P@$$w0rd_#&	**VEC**
EC-009	Completely empty / blank password	(empty)	**IEC**
EC-010	Unregistered credentials for authentication	DummyUnreg99!	**IEC**

EP-03 Feature / Form: Portfolio Backtester (/) Input Field: Portfolio Configuration

EC ID	Equivalence Class Description	Representative Value	Type (VEC / IEC)
EC-011	At least one valid, configured portfolio present	1 portfolio (60/40)	**VEC**
EC-012	Multiple configured comparison portfolios present	2 portfolios (P1, P2)	**VEC**
EC-013	No portfolios configured (0 portfolios present)	0 portfolios	**IEC**
EC-012			

EP-04 Feature / Form: Portfolio Backtester (/) Input Field: Asset Allocation Percentage (%)

EC ID	Equivalence Class Description	Representative Value	Type (VEC / IEC)

EC-014	Total portfolio asset allocations sum to exactly 100%	VTISIM: 60%, BND: 40%	**VEC**
EC-015	Total portfolio asset allocations exceed 100%	VTISIM: 60%, BND: 60% (120%)	**IEC**
EC-016	Total portfolio asset allocations sum to less than 100%	VTISIM: 40%, BND: 40% (80%)	**IEC**
EC-017	Negative asset allocation percentage entered	VTISIM: -10%, BND: 110%	**IEC**

EP-05 Feature / Form: Portfolio Backtester (/) Input Field: Starting Value ($)			
EC ID	**Equivalence Class Description**	**Representative Value**	**Type (VEC / IEC)**
EC-018	Standard positive integer initial investment (>= 1)	$10,000	**VEC**
EC-019	Minimum allowable boundary starting value	$1	**VEC**
EC-020	Zero starting value (strictly < 1)	$0	**IEC**
EC-021	Negative starting value	-$7	**IEC**
EC-022	Completely empty / blank starting value	(empty)	**IEC**

EP-06 Feature / Form: Portfolio Backtester (/) Input Field: Date Range (Start Date vs End Date)			
EC ID	**Equivalence Class Description**	**Representative Value**	**Type (VEC / IEC)**
EC-023	Valid chronological date range (Start < End)	Start: 2010, End: 2023	**VEC**
EC-024	Inverted chronological date range (Start > End)	Start: 2023, End: 2020	**IEC**
EC-025	Identical start and end dates (Start == End)	Start: 2023, End: 2023	**IEC**
EC-026	Start date far in future beyond market dataset	Start: 2099, End: 2100	**IEC**

EP-07 Feature / Form: Portfolio Backtester (/) Input Field: Rolling Window (months)			
EC ID	**Equivalence Class Description**	**Representative Value**	**Type (VEC / IEC)**
EC-027	Standard positive integer duration in months	60 months	**VEC**
EC-028	Zero or negative rolling window duration	0 months / -12 months	**IEC**
EC-029	Empty / blank rolling window	(empty)	**IEC**
EC-028			

EP-08 Feature / Form: Portfolio Backtester (/) Input Field: Cashflows & Rebalancing Frequency			
EC ID	**Equivalence Class Description**	**Representative Value**	**Type (VEC / IEC)**

EC-030	Valid non-negative annual cashflow; standard rebalance	Cashflow: $0, Yearly	**VEC**
EC-031	Valid positive recurring contribution	Cashflow: $500, Monthly	**VEC**
EC-032	Negative recurring cashflow amount	Cashflow: -$500	**IEC**
EC-032			

EP-09 Feature / Form: Monte Carlo Simulator (/monte-carlo) Input Field: Simulation Horizon / Years			
EC ID	**Equivalence Class Description**	**Representative Value**	**Type (VEC / IEC)**
EC-033	Valid positive simulation duration in years	30 years	**VEC**
EC-034	Zero or negative simulation horizon	0 years / -5 years	**IEC**
EC-035	Empty / blank simulation duration	(empty)	**IEC**
EC-036			

EP-10 Feature / Form: Monte Carlo Simulator (/monte-carlo) Input Field: Number of Simulations			
EC ID	**Equivalence Class Description**	**Representative Value**	**Type (VEC / IEC)**
EC-036	Standard positive iteration count (100 to 10,000)	1,000 iterations	**VEC**
EC-037	Zero or negative iteration count	0 / -100	**IEC**
EC-039			
EC-040			

EP-11 Feature / Form: Safe Withdrawal Rate (/safe-withdrawal) Input Field: Withdrawal Rate (%)			
EC ID	**Equivalence Class Description**	**Representative Value**	**Type (VEC / IEC)**
EC-038	Standard sustainable withdrawal percentage (1%–10%)	4.0%	**VEC**
EC-039	Extreme unrealistic withdrawal percentage (> 100%)	120%	**IEC**
EC-040	Negative withdrawal percentage	-3.5%	**IEC**
EC-044			

EP-12 Feature / Form: Backtest Optimizer (/backtest-optimizer) Input Field: Weight Constraints (Min / Max %)			
EC ID	**Equivalence Class Description**	**Representative Value**	**Type (VEC / IEC)**
EC-041	Valid consistent bounds (0% <= Min <= Max <= 100%)	Min: 10%, Max: 50%	**VEC**
EC-042	Inconsistent bounds (Min > Max)	Min: 60%, Max: 40%	**IEC**
EC-043	Negative weight bound	Min: -10%, Max: 50%	**IEC**
EC-048			

EP-13 Feature / Form: Efficient Frontier (/efficient-frontier) Input Field: Asset Selection Count			
EC ID	**Equivalence Class Description**	**Representative Value**	**Type (VEC / IEC)**
EC-044	Multi-asset selection capable of forming curve (>= 2)	3 assets (VTI, BND, GLD)	**VEC**
EC-045	Single asset selection (cannot calculate variance)	1 asset (VTI)	**IEC**
EC-046	Zero assets selected	0 assets	**IEC**
EC-052			

EP-14 Feature / Form: Contact Form (/contact) Input Field: Email			
EC ID	**Equivalence Class Description**	**Representative Value**	**Type (VEC / IEC)**
EC-047	Valid institutional email address	i243137@isb.nu.edu.pk	**VEC**
EC-048	Invalid email format lacking domain structure	user@	**IEC**
EC-049	Empty mandatory email field	(empty)	**IEC**
EC-056			

EP-15 Feature / Form: Contact Form (/contact) Input Field: Subject			
EC ID	**Equivalence Class Description**	**Representative Value**	**Type (VEC / IEC)**
EC-050	Valid subject within length bound (1 to 120 chars)	Inquiry on backtest data	**VEC**
EC-051	Subject exceeding max character limit (> 120 chars)	A * 125 chars	**IEC**
EC-052	Empty subject string	(empty)	**IEC**
EC-060			

EP-16 Feature / Form: Contact Form (/contact) Input Field: Message Body			
EC ID	**Equivalence Class Description**	**Representative Value**	**Type (VEC / IEC)**
EC-053	Valid message body within permitted range (10–5,000)	This is a verified test inquiry.	**VEC**
EC-054	Message body below minimum required length (< 10)	ijklmno (7 chars)	**IEC**
EC-055	Completely empty / blank message body	(empty)	**IEC**
EC-056	Message body exceeding maximum boundary (> 5,000)	X * 5,005 chars	**IEC**

EP-17 Feature / Form: Contact Form (/contact) Input Field: Security Verification (Cloudflare Turnstile)			
EC ID	**Equivalence Class Description**	**Representative Value**	**Type (VEC / IEC)**
EC-057	Cloudflare verification token completed successfully	Success! token verified	**VEC**
EC-058	Unverified / pending / expired captcha token	Missing / unsolved token	**IEC**
EC-067			
EC-068			

EP-18 Feature / Form: Lump Sum vs DCA (/lumpsum-vs-dca) Input Field: Initial Lump Sum ($)			
EC ID	**Equivalence Class Description**	**Representative Value**	**Type (VEC / IEC)**
EC-059	Standard positive initial lump sum amount (>= 1)	$50,000	**VEC**
EC-060	Zero or negative initial investment amount	$0 / -$1,000	**IEC**
EC-061	Empty / blank initial amount	(empty)	**IEC**
EC-072			

EP-19 Feature / Form: Time Value of Money (/time-value-of-money) Input Field: Starting Value ($)			
EC ID	**Equivalence Class Description**	**Representative Value**	**Type (VEC / IEC)**
EC-064	Standard positive initial principal amount (>= 1)	$10,000	**VEC**
EC-065	Zero or negative initial principal	$0 / -$500	**IEC**
EC-066	Empty / blank starting value	(empty)	**IEC**
EC-076			

EP-20 Feature / Form: Time Value of Money (/time-value-of-money) Input Field: Monthly Contribution ($)			
EC ID	**Equivalence Class Description**	**Representative Value**	**Type (VEC / IEC)**
EC-067	Valid monthly contribution in range ($0 to $100,000)	$1,000	**VEC**
EC-068	Exact lower boundary contribution ($0)	$0	**VEC**
EC-069	Negative monthly contribution (< $0)	-$500	**IEC**
EC-070	Excessive monthly contribution (> $100,000)	$150,000	**IEC**
EC-071	Completely empty / blank contribution field	(empty)	**IEC**

LAYER 2 — EP Test Case Table (one combined table — cite EC IDs covered by each test case)

Each TC cites the EC IDs it exercises. Apply Weak EP with Robustness to all forms (one TC per EC, including IECs). Apply Strong EP to your designated critical form. Add rows as needed.

TC-EP ID	EC IDs Covered	Feature / Form	Input Field(s)	Test Value(s)	Expected Result	Actual Result	P/F
TC-EP-001	EC-001, EC-006	Auth Modal (/start)	Email, Password	Email: test@example.com, Password: Testfol123!	Credentials accepted or account created	Pending Execution	**Not Executed**
TC-EP-002	EC-002	Auth Modal (/start)	Email	Email: i243137@isb.nu.edu.pk, Password: ValidPass123!	Educational domain accepted	Pending Execution	**Not Executed**
TC-EP-003	EC-003	Auth Modal (/start)	Email	Email: notanemail, Password: Pass123!	Reject invalid format; show email syntax error	Client-side tooltip: *"Please include an '@' in the email address."*	**Pass**
TC-EP-004	EC-004	Auth Modal (/start)	Email	Email: user@domain, Password: Pass123!	Reject format lacking valid TLD	Tooltip or submission blocked	**Pass**
TC-EP-005	EC-005	Auth Modal (/start)	Email	Email: "", Password: Pass123!	Block empty email submission	Submit button remains disabled or prompts for email	**Pass**
TC-EP-006	EC-007	Auth Modal (/start)	Password	Email: test@example.com, Password: a (1 char)	Check minimum input threshold for UI activation	Submit button activates with 1 char; dispatches request	**Pass**
TC-EP-007	EC-008	Auth Modal (/start)	Password	Email: test@example.com, Password: P@$$w0rd_#&	Accept complex special characters	Password accepted with special symbols	**Pass**
TC-EP-008	EC-009	Auth Modal (/start)	Password	Email: probe_check@test.com, Password: ""	Prevent submission with blank password	Submit button remains dimmed and disabled	**Pass**
TC-EP-009	EC-010	Auth Modal (/login)	Password, Email	Email: test@example.com, Password: • (unregistered)	Authentication fails gracefully with secure notice	Red notification banner: *"Email or password wasn't recognized..."*	**Pass**
TC-EP-010	EC-011	Portfolio Backtester (/)	Portfolios	1 valid portfolio (VTISIM: 60%, BNDSIM: 40%)	Backtest runs and renders metrics	Backtest simulation executes successfully	**Pass**
TC-EP-011	EC-012	Portfolio Backtester (/)	Portfolios	Portfolio 1 (60/40), Portfolio 2 (100% VTISIM)	Multi-portfolio comparison charts generated	Dual portfolio comparison curves render	**Pass**
TC-EP-012	EC-013	Portfolio Backtester (/)	Portfolios	0 portfolios added (empty portfolio list)	Block simulation; alert user to add portfolio	Red banner: *"No valid portfolios were found."*	**Pass**
TC-EP-013	EC-014	Portfolio Backtester (/)	Allocations	VTISIM: 60%, BNDSIM: 40% (Sum = 100%)	Allocation accepted; backtest allowed	Allocations sum to 100; backtest enabled	**Pass**
TC-EP-014	EC-015	Portfolio Backtester (/)	Allocations	VTISIM: 60%, BNDSIM: 60% (Sum = 120%)	Reject allocation sum > 100%	Red banner: *"Portfolio 1: Allocation percentages must sum to 100"*	**Pass**
TC-EP-015	EC-016	Portfolio Backtester (/)	Allocations	VTISIM: 40%, BNDSIM: 40% (Sum = 80%)	Reject allocation sum < 100%	Allocation sum turns red (80%); backtest rejected	**Pass**
TC-EP-016	EC-017	Portfolio Backtester (/)	Allocations	VTISIM: -10%, BNDSIM: 110%	Reject negative allocation weight	Negative sign blocked or rejected on submit	**Pass**

TC-EP ID	EC IDs Covered	Feature / Form	Input Field(s)	Test Value(s)	Expected Result	Actual Result	P/F
TC-EP-017	EC-018	Portfolio Backtester (/)	Starting Value	Starting Value: $10,000	Standard initial investment accepted	Pending Execution	**Not Executed**
TC-EP-018	EC-019	Portfolio Backtester (/)	Starting Value	Starting Value: $1	Minimum valid integer accepted	Pending Execution	**Not Executed**
TC-EP-019	EC-020	Portfolio Backtester (/)	Starting Value	Starting Value: $0	Reject zero starting value	Pending Execution	**Not Executed**
TC-EP-020	EC-021	Portfolio Backtester (/)	Starting Value	Starting Value: -$7	Reject negative initial cash	Tooltip: *"Value must be greater than or equal to 1."*	**Pass**
TC-EP-021	EC-022	Portfolio Backtester (/)	Starting Value	Starting Value: "" (empty)	Reject empty initial investment	Red banner: *"Invalid starting value"*	**Pass**
TC-EP-022	EC-023	Portfolio Backtester (/)	Dates	Start: 01/01/2010, End: 12/31/2023	Valid chronological period accepted	Pending Execution	**Not Executed**
TC-EP-023	EC-024	Portfolio Backtester (/)	Dates	Start: 01/01/2023, End: 01/01/2020	Reject inverted chronological dates	Red banner: *"Start date must be before end date"*	**Pass**
TC-EP-024	EC-025	Portfolio Backtester (/)	Dates	Start: 01/01/2023, End: 01/01/2023	Reject zero-duration time span	Red banner: *"Start date must be before end date"*	**Pass**
TC-EP-025	EC-026	Portfolio Backtester (/)	Dates	Start: 01/01/2099, End: 01/01/2100	Trap future date without market data	Notice: No historical prices for selected period	**Pass**
TC-EP-026	EC-027	Portfolio Backtester (/)	Rolling Window	Rolling Window: 60 months	Valid rolling metric period accepted	Pending Execution	**Not Executed**
TC-EP-027	EC-028	Portfolio Backtester (/)	Rolling Window	Rolling Window: 0 or -12	Reject non-positive window duration	Field reset or tooltip prompting integer >= 1	**Pass**
TC-EP-028	EC-029	Portfolio Backtester (/)	Rolling Window	Rolling Window: "" (empty)	Reject missing window parameter	Prompt to enter valid rolling window	**Pass**
TC-EP-029	EC-030	Portfolio Backtester (/)	Cashflows	Cashflow: $0, Frequency: Yearly	Zero recurring cashflow treated as lump sum	Pending Execution	**Not Executed**
TC-EP-030	EC-031	Portfolio Backtester (/)	Cashflows	Cashflow: $500, Frequency: Monthly	Periodic deposits modeled correctly	Growth reflects $500/mo cashflow additions	**Pass**
TC-EP-031	EC-032	Portfolio Backtester (/)	Cashflows	Cashflow: -$500	Reject negative contribution (or treat as withdrawal)	Treated as scheduled withdrawal leg	**Pass**
TC-EP-032	EC-033	Monte Carlo (/monte-carlo)	Horizon	Years: 30	Valid 30-year simulation computed	Blocked by tier restriction: Free tier caps horizon at 15 yrs; triggers upgrade modal	**Blocked**
TC-EP-033	EC-034	Monte Carlo (/monte-carlo)	Horizon	Years: 0	Reject zero simulation horizon	Error / warning: horizon must be at least 1 yr	**Pass**
TC-EP-034	EC-035	Monte Carlo (/monte-carlo)	Horizon	Years: "" (empty)	Reject missing horizon	Validation prompt for duration	**Pass**
TC-EP-035	EC-036	Monte Carlo (/monte-carlo)	Simulations	Number of Simulations: 1,000	Standard simulation count executes	Pending Execution	**Not Executed**
TC-EP-036	EC-037	Monte Carlo (/monte-carlo)	Simulations	Number of Simulations: 0	Reject zero simulations	Pending Execution	**Not Executed**
TC-EP-037	EC-038	Safe Withdrawal (/safe-withdrawal-rate-calculator)	Rate (%)	Withdrawal Rate: 4.0%	Trinity study baseline simulated	Safe failure probability computed	**Pass**
TC-EP-038	EC-039	Safe Withdrawal (/safe-withdrawal-rate-calculator)	Rate (%)	Withdrawal Rate: 120%	Reject or warn of immediate portfolio depletion	Depletion warning or validation bounds	**Pass**

TC-EP ID	EC IDs Covered	Feature / Form	Input Field(s)	Test Value(s)	Expected Result	Actual Result	P/F
TC-EP-039	EC-040	Safe Withdrawal (/safe-withdrawal-rate-calculator)	Rate (%)	Withdrawal Rate: -3.5%	Reject negative withdrawal rate	Negative rate blocked by input mask	**Pass**
TC-EP-040	EC-041	Backtest Optimizer (/backtest-optimizer)	Weight Bounds	Min: 10%, Max: 50%	Optimizer solves constrained weights	Optimal asset weights bounded in [10%, 50%]	**Pass**
TC-EP-041	EC-042	Backtest Optimizer (/backtest-optimizer)	Weight Bounds	Min: 60%, Max: 40%	Reject inconsistent bounds (Min > Max)	Optimizer error: Min bound cannot exceed Max	**Pass**
TC-EP-042	EC-043	Backtest Optimizer (/backtest-optimizer)	Weight Bounds	Min: -10%	Reject negative bounds (shorting disallowed)	Field resets or shorting disallowed error	**Pass**
TC-EP-043	EC-044	Efficient Frontier (/efficient-frontier)	Assets	3 assets (VTI, BND, GLD)	Compute efficient covariance frontier	Curved frontier plot renders	**Pass**
TC-EP-044	EC-045	Efficient Frontier (/efficient-frontier)	Assets	1 asset (VTI)	Reject single asset; frontier requires portfolio	Error: At least 2 assets needed for frontier	**Pass**
TC-EP-045	EC-046	Efficient Frontier (/efficient-frontier)	Assets	0 assets selected	Block execution	Pending Execution	**Not Executed**
TC-EP-046	EC-047	Contact Form (/contact)	Email	Email: i243137@isb.nu.edu.pk	Valid institutional email accepted	Pending Execution	**Not Executed**
TC-EP-047	EC-048	Contact Form (/contact)	Email	Email: user@	Reject incomplete email address	Pending Execution	**Not Executed**
TC-EP-048	EC-049	Contact Form (/contact)	Email	Email: ""	Reject missing required email	Pending Execution	**Not Executed**
TC-EP-049	EC-050	Contact Form (/contact)	Subject	Subject: Inquiry on backtest data	Valid subject accepted	Pending Execution	**Not Executed**
TC-EP-050	EC-051	Contact Form (/contact)	Subject	Subject: 125 chars	Enforce 120 char max boundary	Pending Execution	**Not Executed**
TC-EP-051	EC-052	Contact Form (/contact)	Subject	Subject: "" (empty)	Allow or prompt for optional subject	Pending Execution	**Not Executed**
TC-EP-052	EC-053, EC-057	Contact Form (/contact)	Message, Captcha	Message: 32 chars, Captcha: Verified	Message sent successfully to support inbox	Green banner: *"Message sent. We will get back to you soon."*	**Pass**
TC-EP-053	EC-054	Contact Form (/contact)	Message	Message: ijklmno (7 chars)	Reject message < 10 characters	Red banner: *"Message must be at least 10 characters."*	**Pass**
TC-EP-054	EC-055	Contact Form (/contact)	Message	Message: "" (empty)	Reject blank message	Pending Execution	**Not Executed**
TC-EP-055	EC-056	Contact Form (/contact)	Message	Message: 5,005 chars	Reject message exceeding 5,000 characters	Pending Execution	**Not Executed**
TC-EP-056	EC-058	Contact Form (/contact)	Captcha	Message: valid, Captcha: Unsolved	Block submission until anti-abuse check passes	Pending Execution	**Not Executed**
TC-EP-057	EC-059	Lump Sum vs DCA (/lumpsum-vs-dca)	Initial Lump Sum	Lump Sum: $50,000	Valid lump sum comparative backtest runs	Blocked by paywall: Free users redirected to /pricing table ("Recommended for Lump Sum vs DCA")	**Blocked**
TC-EP-058	EC-060	Lump Sum vs DCA (/lumpsum-vs-dca)	Initial Lump Sum	Lump Sum: $0	Reject zero initial lump sum	Pending Execution	**Not Executed**
TC-EP-059	EC-061	Lump Sum vs DCA (/lumpsum-vs-dca)	Initial Lump Sum	Lump Sum: ""	Reject missing initial cash	Pending Execution	**Not Executed**
TC-EP-060	EC-062	Lump Sum vs DCA (/lumpsum-vs-dca)	DCA Contribution	DCA Monthly: $1,000	Valid periodic contribution modeled	Pending Execution	**Not Executed**
TC-EP-061	EC-063	Lump Sum vs DCA (/lumpsum-vs-dca)	DCA Contribution	DCA Monthly: -$200	Reject negative contribution amount	Pending Execution	**Not Executed**

TC-EP ID	EC IDs Covered	Feature / Form	Input Field(s)	Test Value(s)	Expected Result	Actual Result	P/F
TC-EP-062	EC-064	TVM Calculator (/time-value-of-money)	Starting Value	Starting Value: $10,000	Valid present investment computed	Blocked by access control: Gated tool triggers "Sign in or create your account" modal	**Blocked**
TC-EP-063	EC-065	TVM Calculator (/time-value-of-money)	Starting Value	Starting Value: $0	Handle zero initial wealth	Pending Execution	**Not Executed**
TC-EP-064	EC-066	TVM Calculator (/time-value-of-money)	Starting Value	Starting Value: ""	Reject missing starting value	Pending Execution	**Not Executed**
TC-EP-065	EC-067	TVM Calculator (/time-value-of-money)	Contribution	Monthly Contribution: $1,000	Valid periodic contribution computed	Results sampled and plotted	**Pass**
TC-EP-066	EC-068	TVM Calculator (/time-value-of-money)	Contribution	Monthly Contribution: $0	Exact lower bound accepted	Successfully renders chart: *"Results (sampled 1805 feasible solutions)"*	**Pass**
TC-EP-067	EC-069	TVM Calculator (/time-value-of-money)	Contribution	Monthly Contribution: -$500	Reject negative contribution amount	Red banner: *"Contribution must be between $0 and $100,000."*	**Pass**
TC-EP-068	EC-070	TVM Calculator (/time-value-of-money)	Contribution	Monthly Contribution: $150,000	Reject contribution > $100,000	Pending Execution	**Not Executed**
TC-EP-069	EC-071	TVM Calculator (/time-value-of-money)	Contribution	Monthly Contribution: ""	Reject empty contribution	Banner: *"Contribution must be between $0 and $100,000."*	**Pass**
TC-EP-070	EC-072	Rebalancing Sensitivity (/rebalancing-sensitivity)	Frequency	Frequency: Monthly	Models rebalancing drag on monthly basis	Sensitivity plot generated	**Pass**
TC-EP-071	EC-073	Rebalancing Sensitivity (/rebalancing-sensitivity)	Frequency	Frequency: Never	Models drift without rebalancing	Pending Execution	**Not Executed**
TC-EP-072	EC-074	Rebalancing Sensitivity (/rebalancing-sensitivity)	Frequency	Frequency: null	Highlight mandatory selection	Pending Execution	**Not Executed**
TC-EP-073	EC-011, EC-014, EC-018, EC-023, EC-027, EC-030	Portfolio Backtester	Portfolios, Allocations, Starting Value, Dates, Rolling Window, Cashflow	Portfolio: 1 present (VTISIM: 60%, BNDSIM: 40%), Starting Value: $10,000, Start: 01/01/2015, End: 12/31/2023, Rolling: 60, Cashflow: $0	All Valid Combinations: Entire backtest suite executes, generating full performance summary, drawdowns, and Sharpe ratio	Pending Execution	**Not Executed**
TC-EP-074	EC-011, EC-015, EC-018, EC-023	Portfolio Backtester	Portfolios, Allocations, Starting Value, Dates	Valid Portfolio, Allocations: 60% + 60% = 120% (Invalid), Starting Value: $10,000 (Valid), Start: 2015, End: 2023 (Valid)	Multi-variable with Invalid Allocation: Allocation error halts submission despite all other fields being valid	Pending Execution	**Not Executed**
TC-EP-075	EC-011, EC-014, EC-021, EC-023	Portfolio Backtester	Portfolios, Allocations, Starting Value, Dates	Valid Portfolio, Allocations: 60/40 (Valid), Starting Value: -$7 (Invalid), Dates: 2015–2023 (Valid)	Multi-variable with Invalid Starting Value: Tooltip flags starting cash error despite valid allocations and dates	Pending Execution	**Not Executed**

TC-EP ID	EC IDs Covered	Feature / Form	Input Field(s)	Test Value(s)	Expected Result	Actual Result	P/F
TC-EP-076	EC-011, EC-014, EC-018, EC-024	Portfolio Backtester	Portfolios, Allocations, Starting Value, Dates	Valid Portfolio, Allocations: 60/40 (Valid), Starting Value: $10,000 (Valid), Start: 2023, End: 2020 (Invalid)	Multi-variable with Inverted Date Range: Date error halts submission despite valid portfolio and starting value	Pending Execution	**Not Executed**
TC-EP-077	EC-013, EC-015, EC-020, EC-024	Portfolio Backtester	Portfolios, Allocations, Starting Value, Dates	0 Portfolios (Invalid), Allocations: 120% (Invalid), Starting Value: $0 (Invalid), Start > End (Invalid)	Extreme Multi-Failure Robustness: Tests prioritization of error trapping when all major subsystems are in invalid state	Pending Execution	**Not Executed**

TOTAL EP TEST CASES:	**77**

D3 — Boundary Value Analysis [25 marks]

ℹ *D3 has two layers. Complete Layer 1 first — one boundary table per input field where BVA is applicable (your decision). Then write Layer 2 — your BVA test cases, citing the BC IDs from Layer 1.*

LAYER 1 — BVA Boundary Tables (one table per applicable input field — identify boundaries and assign BC IDs)

ℹ *Add a boundary table only for fields where BVA applies. Assign BC IDs sequentially (BC-001, BC-002, …) across all fields. Add more tables as needed.*

BV-01

BC ID	Feature / Form	Input Field	EC ID	Boundary Description	Boundary Value (inferred from probing)
BC-001	Auth Modal (/start)	Password	**EC-007**	Minimum password length required to activate submit button (lower bound)	**1 character**
BC-002				*e.g. Maximum password length (upper boundary)*	

BV-02

BC ID	Feature / Form	Input Field	EC ID	Boundary Description	Boundary Value (inferred from probing)
BC-002	Portfolio Backtester (/)	Allocation Sum	**EC-014**	Lower exact bound for total portfolio asset allocation percentage	**100%**
BC-004					

BV-03

BC ID	Feature / Form	Input Field	EC ID	Boundary Description	Boundary Value (inferred from probing)
BC-003	Portfolio Backtester (/)	Allocation Sum	**EC-014**	Upper exact bound for total portfolio asset allocation percentage	**100%**
BC-006					

BV-04					
BC ID	**Feature / Form**	**Input Field**	**EC ID**	**Boundary Description**	**Boundary Value (inferred from probing)**
BC-004	Portfolio Backtester (/)	Starting Value ($)	**EC-019**	Absolute minimum allowable initial starting capital (lower bound)	**$1**
BC-008					

BV-05					
BC ID	**Feature / Form**	**Input Field**	**EC ID**	**Boundary Description**	**Boundary Value (inferred from probing)**
BC-005	Portfolio Backtester (/)	Date Range	**EC-023**	Relational chronological threshold: Start Date must be strictly prior to End Date	**Start = End - 1 day**
BC-010					

BV-06					
BC ID	**Feature / Form**	**Input Field**	**EC ID**	**Boundary Description**	**Boundary Value (inferred from probing)**
BC-006	Portfolio Backtester (/)	Rolling Window	**EC-027**	Minimum duration for calculating rolling performance metrics (lower bound)	**1 month**
BC-012					

BV-07					
BC ID	**Feature / Form**	**Input Field**	**EC ID**	**Boundary Description**	**Boundary Value (inferred from probing)**
BC-007	Monte Carlo (/monte-carlo)	Simulation Horizon	**EC-033**	Minimum time horizon for stochastic multi-year path modeling (lower bound)	**1 year**
BC-014					

BV-08

BC ID	Feature / Form	Input Field	EC ID	Boundary Description	Boundary Value (inferred from probing)
BC-008	Safe Withdrawal (/safe-withdrawal)	Withdrawal Rate (%)	**EC-038**	Minimum allowable non-negative withdrawal percentage (lower bound)	**0.0%**
BC-016					

BV-09					
BC ID	**Feature / Form**	**Input Field**	**EC ID**	**Boundary Description**	**Boundary Value (inferred from probing)**
BC-009	Contact Form (/contact)	Message Length	**EC-054**	Minimum required character length for message inquiry body (lower bound)	**10 characters**
BC-018					

BV-10					
BC ID	**Feature / Form**	**Input Field**	**EC ID**	**Boundary Description**	**Boundary Value (inferred from probing)**
BC-010	Contact Form (/contact)	Message Length	**EC-056**	Maximum allowable character length for message inquiry body (upper bound)	**5,000 characters**
BC-020					

BV-11					
BC ID	**Feature / Form**	**Input Field**	**EC ID**	**Boundary Description**	**Boundary Value (inferred from probing)**
BC-011	Contact Form (/contact)	Subject Length	**EC-051**	Maximum allowable character length for inquiry subject line (upper bound)	**120 characters**
BC-022					

BV-12

BC ID	Feature / Form	Input Field	EC ID	Boundary Description	Boundary Value (inferred from probing)
BC-012	Lump Sum vs DCA (/lumpsum-vs-dca)	Initial Lump Sum ($)	**EC-059**	Minimum initial lump sum amount for comparative analysis (lower bound)	**$1**
BC-024					

BV-13					
BC ID	**Feature / Form**	**Input Field**	**EC ID**	**Boundary Description**	**Boundary Value (inferred from probing)**
BC-013	Time Value of Money (/time-value-of-money)	Monthly Contribution ($)	**EC-068**	Minimum allowable periodic contribution amount (lower bound)	**$0**
BC-026					

BV-14					
BC ID	**Feature / Form**	**Input Field**	**EC ID**	**Boundary Description**	**Boundary Value (inferred from probing)**
BC-014	Time Value of Money (/time-value-of-money)	Monthly Contribution ($)	**EC-070**	Maximum allowable periodic contribution amount (upper bound)	**$100,000**
BC-028					

BV-15					
BC ID	**Feature / Form**	**Input Field**	**EC ID**	**Boundary Description**	**Boundary Value (inferred from probing)**
BC-015	Time Value of Money (/time-value-of-money)	Starting Value ($)	**EC-064**	Minimum initial present capital for TVM calculation (lower bound)	**$1**
BC-030					

LAYER 2 — BVA Test Case Table (one combined table — cite BC IDs and state position: Below / At / Above boundary)

Each TC cites the BC ID it probes and states the test value's position relative to the boundary. Apply 2-point BVA to all forms. Apply 3-point BVA to your designated critical form. Add rows as needed.

TC-BVA ID	BC IDs	Feature / Form	Input Field	Test Value	Position	Expected Result	Actual Result	P/F
TC-BVA-001	BC-002, BC-003	Portfolio Backtester	Allocation Sum (%)	80% (VTISIM: 40%, BNDSIM: 40%)	**Below**	Reject under-allocated portfolio	Allocation sum highlights red (80%); red banner: *"Allocations must sum to 100"*	**Pass**
TC-BVA-002	BC-002, BC-003	Portfolio Backtester	Allocation Sum (%)	100% (VTISIM: 60%, BNDSIM: 40%)	**At**	Accept exact 100% allocation	Backtest executes simulation successfully	**Pass**
TC-BVA-003	BC-002, BC-003	Portfolio Backtester	Allocation Sum (%)	120% (VTISIM: 60%, BNDSIM: 60%)	**Above**	Reject over-allocated portfolio	Red banner: *"Portfolio 1: Allocation percentages must sum to 100"*	**Pass**
TC-BVA-004	BC-004	Portfolio Backtester	Starting Value ($)	$0	**Below**	Reject non-positive starting cash	Tooltip: "Value must be greater than or equal to 1."	**Pass**
TC-BVA-005	BC-004	Portfolio Backtester	Starting Value ($)	$1	**At**	Accept exact minimum allowable investment	Pending Execution	**Not Executed**
TC-BVA-006	BC-004	Portfolio Backtester	Starting Value ($)	$10,000	**Above**	Accept positive investment above minimum	Backtest executes simulation successfully ("Backtest complete")	**Pass**
TC-BVA-007	BC-005	Portfolio Backtester	Date Range	Start: 01/01/2023, End: 01/01/2023	**At (Diff = 0)**	Reject zero duration interval	Red banner: "Start date must be before end date"	**Pass**
TC-BVA-008	BC-005	Portfolio Backtester	Date Range	Start: 01/01/2023, End: 01/01/2020	**Below (Start > End)**	Reject inverted date interval	Red banner: *"Start date must be before end date"*	**Pass**
TC-BVA-009	BC-005	Portfolio Backtester	Date Range	Start: 01/01/2025, End: 10/26/2026	**Above (Start < End)**	Accept valid chronological date interval	Backtest accepts range and computes successfully	**Pass**
TC-BVA-010	BC-006	Portfolio Backtester	Rolling Window	0 months	**Below**	Reject zero rolling window	Tooltip: "Value must be greater than or equal to 1."	**Pass**
TC-BVA-011	BC-006	Portfolio Backtester	Rolling Window	1 month	**At**	Accept minimum 1-month rolling calculation	Pending Execution	**Not Executed**
TC-BVA-012	BC-006	Portfolio Backtester	Rolling Window	60 months	**Above**	Accept valid rolling calculation above minimum	Backtest executes simulation successfully	**Pass**
TC-BVA-013	BC-001	Auth Modal (/start)	Password Length	0 chars ("")	**Below**	Submit button disabled	Submit button dimmed / disabled client-side	**Pass**
TC-BVA-014	BC-001	Auth Modal (/start)	Password Length	1 char ("a")	**At**	Submit button activates	Submit button enabled client-side	**Pass**

TC-BVA ID	BC IDs	Feature / Form	Input Field	Test Value	Position	Expected Result	Actual Result	P/F
TC-BVA-015	BC-007	Monte Carlo (/monte-carlo)	Horizon (Years)	0 years	**Below**	Reject zero horizon	Red banner: "Number of years must be positive"	**Pass**
TC-BVA-016	BC-007	Monte Carlo (/monte-carlo)	Horizon (Years)	1 year	**At**	Accept minimum 1-year stochastic run	Red banner: "Phase 1 withdrawal rate must be between 0% and 100%"	**Pass**
TC-BVA-017	BC-008	Safe Withdrawal (/safe-withdrawal)	Withdrawal Rate (%)	-0.1%	**Below**	Reject negative withdrawal percentage	Red banner: "Phase 1 withdrawal rate must be between 0% and 100%"	**Pass**
TC-BVA-018	BC-008	Safe Withdrawal (/safe-withdrawal)	Withdrawal Rate (%)	0.0%	**At**	Accept zero withdrawal rate (capital preservation)	Red banner: "Phase 1 withdrawal rate must be between 0% and 100%"	**Pass**
TC-BVA-019	BC-009	Contact Form (/contact)	Message Length	9 chars ("123456789")	**Below**	Reject message under 10 chars	Red banner: "Message must be at least 10 characters."	**Pass**
TC-BVA-020	BC-009	Contact Form (/contact)	Message Length	10 chars ("1234567890")	**At**	Accept exact minimum 10 char message	Pending Execution	**Not Executed**
TC-BVA-021	BC-010	Contact Form (/contact)	Message Length	5,000 chars	**At**	Accept exact maximum 5,000 char message	Pending Execution	**Not Executed**
TC-BVA-022	BC-010	Contact Form (/contact)	Message Length	5,001 chars	**Above**	Reject message exceeding 5,000 chars	Pending Execution	**Not Executed**
TC-BVA-023	BC-011	Contact Form (/contact)	Subject Length	120 chars	**At**	Accept exact maximum 120 char subject	Pending Execution	**Not Executed**
TC-BVA-024	BC-011	Contact Form (/contact)	Subject Length	121 chars	**Above**	Block input exceeding 120 chars	Pending Execution	**Not Executed**
TC-BVA-025	BC-012	Lump Sum vs DCA (/lumpsum-vs-dca)	Initial Lump Sum ($)	$0	**Below**	Reject zero initial investment	Pending Execution	**Not Executed**
TC-BVA-026	BC-012	Lump Sum vs DCA (/lumpsum-vs-dca)	Initial Lump Sum ($)	$1	**At**	Accept minimum $1 starting lump sum	Pending Execution	**Not Executed**
TC-BVA-027	BC-013	TVM Calculator (/time-value-of-money)	Monthly Contribution ($)	-$1	**Below**	Reject negative periodic contribution	Pending Execution	**Not Executed**
TC-BVA-028	BC-013	TVM Calculator (/time-value-of-money)	Monthly Contribution ($)	$0	**At**	Accept exact minimum contribution ($0)	Pending Execution	**Not Executed**
TC-BVA-029	BC-014	TVM Calculator (/time-value-of-money)	Monthly Contribution ($)	$100,000	**At**	Accept exact upper limit contribution	Pending Execution	**Not Executed**
TC-BVA-030	BC-014	TVM Calculator (/time-value-of-money)	Monthly Contribution ($)	$100,001	**Above**	Reject contribution exceeding $100k cap	Pending Execution	**Not Executed**
TC-BVA-031	BC-015	TVM Calculator (/time-value-of-money)	Starting Value ($)	$0	**Below**	Handle zero initial wealth	Pending Execution	**Not Executed**
TC-BVA-032	BC-015	TVM Calculator (/time-value-of-money)	Starting Value ($)	$1	**At**	Accept minimum $1 initial wealth	Pending Execution	**Not Executed**

TOTAL BVA TEST CASES:	**32**

D4 — Traceability Matrix [10 marks]

List ALL test cases from D2 and D3. Every TC must appear here. Add rows as needed.

TC ID	Feature / Form	Input Field	Technique	EC / BC IDs Covered	Status	Result
TC-EP-001	Auth Modal (/start)	Email, Password	EP	EC-001, EC-006	Not Executed	**Pending**
TC-EP-002	Auth Modal (/start)	Email	EP	EC-002	Not Executed	**Pending**
TC-EP-003	Auth Modal (/start)	Email	EP	EC-003	Executed	**Pass**
TC-EP-004	Auth Modal (/start)	Email	EP	EC-004	Executed	**Pass**
TC-EP-005	Auth Modal (/start)	Email	EP	EC-005	Executed	**Pass**
TC-EP-006	Auth Modal (/start)	Password	EP	EC-007	Executed	**Pass**
TC-EP-007	Auth Modal (/start)	Password	EP	EC-008	Executed	**Pass**
TC-EP-008	Auth Modal (/start)	Password	EP	EC-009	Executed	**Pass**
TC-EP-009	Auth Modal (/login)	Password, Email	EP	EC-010	Executed	**Pass**
TC-EP-010	Portfolio Backtester (/)	Portfolios	EP	EC-011	Executed	**Pass**
TC-EP-011	Portfolio Backtester (/)	Portfolios	EP	EC-012	Executed	**Pass**
TC-EP-012	Portfolio Backtester (/)	Portfolios	EP	EC-013	Executed	**Pass**
TC-EP-013	Portfolio Backtester (/)	Allocations	EP	EC-014	Executed	**Pass**
TC-EP-014	Portfolio Backtester (/)	Allocations	EP	EC-015	Executed	**Pass**
TC-EP-015	Portfolio Backtester (/)	Allocations	EP	EC-016	Executed	**Pass**
TC-EP-016	Portfolio Backtester (/)	Allocations	EP	EC-017	Executed	**Pass**
TC-EP-017	Portfolio Backtester (/)	Starting Value	EP	EC-018	Not Executed	**Pending**
TC-EP-018	Portfolio Backtester (/)	Starting Value	EP	EC-019	Not Executed	**Pending**
TC-EP-019	Portfolio Backtester (/)	Starting Value	EP	EC-020	Not Executed	**Pending**
TC-EP-020	Portfolio Backtester (/)	Starting Value	EP	EC-021	Executed	**Pass**
TC-EP-021	Portfolio Backtester (/)	Starting Value	EP	EC-022	Executed	**Pass**
TC-EP-022	Portfolio Backtester (/)	Dates	EP	EC-023	Not Executed	**Pending**
TC-EP-023	Portfolio Backtester (/)	Dates	EP	EC-024	Executed	**Pass**
TC-EP-024	Portfolio Backtester (/)	Dates	EP	EC-025	Executed	**Pass**
TC-EP-025	Portfolio Backtester (/)	Dates	EP	EC-026	Executed	**Pass**
TC-EP-026	Portfolio Backtester (/)	Rolling Window	EP	EC-027	Not Executed	**Pending**
TC-EP-027	Portfolio Backtester (/)	Rolling Window	EP	EC-028	Executed	**Pass**
TC-EP-028	Portfolio Backtester (/)	Rolling Window	EP	EC-029	Executed	**Pass**
TC-EP-029	Portfolio Backtester (/)	Cashflows	EP	EC-030	Not Executed	**Pending**

TC ID	Feature / Form	Input Field	Technique	EC / BC IDs Covered	Status	Result
TC-EP-030	Portfolio Backtester (/)	Cashflows	EP	EC-031	Executed	**Pass**
TC-EP-031	Portfolio Backtester (/)	Cashflows	EP	EC-032	Executed	**Pass**
TC-EP-032	Monte Carlo (/monte-carlo)	Horizon	EP	EC-033	Blocked	**Blocked**
TC-EP-033	Monte Carlo (/monte-carlo)	Horizon	EP	EC-034	Executed	**Pass**
TC-EP-034	Monte Carlo (/monte-carlo)	Horizon	EP	EC-035	Executed	**Pass**
TC-EP-035	Monte Carlo (/monte-carlo)	Simulations	EP	EC-036	Not Executed	**Pending**
TC-EP-036	Monte Carlo (/monte-carlo)	Simulations	EP	EC-037	Not Executed	**Pending**
TC-EP-037	Safe Withdrawal (/safe-withdrawal-rate-calculator)	Rate (%)	EP	EC-038	Executed	**Pass**
TC-EP-038	Safe Withdrawal (/safe-withdrawal-rate-calculator)	Rate (%)	EP	EC-039	Executed	**Pass**
TC-EP-039	Safe Withdrawal (/safe-withdrawal-rate-calculator)	Rate (%)	EP	EC-040	Executed	**Pass**
TC-EP-040	Backtest Optimizer (/backtest-optimizer)	Weight Bounds	EP	EC-041	Executed	**Pass**
TC-EP-041	Backtest Optimizer (/backtest-optimizer)	Weight Bounds	EP	EC-042	Executed	**Pass**
TC-EP-042	Backtest Optimizer (/backtest-optimizer)	Weight Bounds	EP	EC-043	Executed	**Pass**
TC-EP-043	Efficient Frontier (/efficient-frontier)	Assets	EP	EC-044	Executed	**Pass**
TC-EP-044	Efficient Frontier (/efficient-frontier)	Assets	EP	EC-045	Executed	**Pass**
TC-EP-045	Efficient Frontier (/efficient-frontier)	Assets	EP	EC-046	Not Executed	**Pending**
TC-EP-046	Contact Form (/contact)	Email	EP	EC-047	Not Executed	**Pending**
TC-EP-047	Contact Form (/contact)	Email	EP	EC-048	Not Executed	**Pending**
TC-EP-048	Contact Form (/contact)	Email	EP	EC-049	Not Executed	**Pending**
TC-EP-049	Contact Form (/contact)	Subject	EP	EC-050	Not Executed	**Pending**
TC-EP-050	Contact Form (/contact)	Subject	EP	EC-051	Not Executed	**Pending**
TC-EP-051	Contact Form (/contact)	Subject	EP	EC-052	Not Executed	**Pending**
TC-EP-052	Contact Form (/contact)	Message, Captcha	EP	EC-053, EC-057	Executed	**Pass**
TC-EP-053	Contact Form (/contact)	Message	EP	EC-054	Executed	**Pass**
TC-EP-054	Contact Form (/contact)	Message	EP	EC-055	Not Executed	**Pending**
TC-EP-055	Contact Form (/contact)	Message	EP	EC-056	Not Executed	**Pending**
TC-EP-056	Contact Form (/contact)	Captcha	EP	EC-058	Not Executed	**Pending**

TC ID	Feature / Form	Input Field	Technique	EC / BC IDs Covered	Status	Result
TC-EP-057	Lump Sum vs DCA (/lumpsum-vs-dca)	Initial Lump Sum	EP	EC-059	Blocked	**Blocked**
TC-EP-058	Lump Sum vs DCA (/lumpsum-vs-dca)	Initial Lump Sum	EP	EC-060	Not Executed	**Pending**
TC-EP-059	Lump Sum vs DCA (/lumpsum-vs-dca)	Initial Lump Sum	EP	EC-061	Not Executed	**Pending**
TC-EP-060	Lump Sum vs DCA (/lumpsum-vs-dca)	DCA Contribution	EP	EC-062	Not Executed	**Pending**
TC-EP-061	Lump Sum vs DCA (/lumpsum-vs-dca)	DCA Contribution	EP	EC-063	Not Executed	**Pending**
TC-EP-062	TVM Calculator (/time-value-of-money)	Starting Value	EP	EC-064	Blocked	**Blocked**
TC-EP-063	TVM Calculator (/time-value-of-money)	Starting Value	EP	EC-065	Not Executed	**Pending**
TC-EP-064	TVM Calculator (/time-value-of-money)	Starting Value	EP	EC-066	Not Executed	**Pending**
TC-EP-065	TVM Calculator (/time-value-of-money)	Contribution	EP	EC-067	Executed	**Pass**
TC-EP-066	TVM Calculator (/time-value-of-money)	Contribution	EP	EC-068	Executed	**Pass**
TC-EP-067	TVM Calculator (/time-value-of-money)	Contribution	EP	EC-069	Executed	**Pass**
TC-EP-068	TVM Calculator (/time-value-of-money)	Contribution	EP	EC-070	Not Executed	**Pending**
TC-EP-069	TVM Calculator (/time-value-of-money)	Contribution	EP	EC-071	Executed	**Pass**
TC-EP-070	Rebalancing Sensitivity (/rebalancing-sensitivity)	Frequency	EP	EC-072	Blocked	**Blocked**
TC-EP-071	Rebalancing Sensitivity (/rebalancing-sensitivity)	Frequency	EP	EC-073	Not Executed	**Pending**
TC-EP-072	Rebalancing Sensitivity (/rebalancing-sensitivity)	Frequency	EP	EC-074	Not Executed	**Pending**
TC-EP-073	Portfolio Backtester	Portfolios, Allocations, Starting Value, Dates, Rolling Window, Cashflow	EP	EC-011, EC-014, EC-018, EC-023, EC-027, EC-030	Not Executed	**Pending**
TC-EP-074	Portfolio Backtester	Portfolios, Allocations, Starting Value, Dates	EP	EC-011, EC-015, EC-018, EC-023	Not Executed	**Pending**
TC-EP-075	Portfolio Backtester	Portfolios, Allocations, Starting Value, Dates	EP	EC-011, EC-014, EC-021, EC-023	Not Executed	**Pending**
TC-EP-076	Portfolio Backtester	Portfolios, Allocations, Starting Value, Dates	EP	EC-011, EC-014, EC-018, EC-024	Not Executed	**Pending**
TC-EP-077	Portfolio Backtester	Portfolios, Allocations, Starting Value, Dates	EP	EC-013, EC-015, EC-020, EC-024	Not Executed	**Pending**

TC ID	Feature / Form	Input Field	Technique	EC / BC IDs Covered	Status	Result
TC-BVA-001	Portfolio Backtester	Allocation Sum (%)	BVA	BC-002, BC-003 (Below)	Executed	**Pass**
TC-BVA-002	Portfolio Backtester	Allocation Sum (%)	BVA	BC-002, BC-003 (At)	Executed	**Pass**
TC-BVA-003	Portfolio Backtester	Allocation Sum (%)	BVA	BC-002, BC-003 (Above)	Executed	**Pass**
TC-BVA-004	Portfolio Backtester	Starting Value ($)	BVA	BC-004 (Below)	Executed	**Pass**
TC-BVA-005	Portfolio Backtester	Starting Value ($)	BVA	BC-004 (At)	Not Executed	**Pending**
TC-BVA-006	Portfolio Backtester	Starting Value ($)	BVA	BC-004 (Above)	Executed	**Pass**
TC-BVA-007	Portfolio Backtester	Date Range	BVA	BC-005 (At (Diff = 0))	Executed	**Pass**
TC-BVA-008	Portfolio Backtester	Date Range	BVA	BC-005 (Below (Start > End))	Executed	**Pass**
TC-BVA-009	Portfolio Backtester	Date Range	BVA	BC-005 (Above (Start < End by 1 day))	Executed	**Pass**
TC-BVA-010	Portfolio Backtester	Rolling Window	BVA	BC-006 (Below)	Executed	**Pass**
TC-BVA-011	Portfolio Backtester	Rolling Window	BVA	BC-006 (At)	Not Executed	**Pending**
TC-BVA-012	Portfolio Backtester	Rolling Window	BVA	BC-006 (Above)	Executed	**Pass**
TC-BVA-013	Auth Modal (/start)	Password Length	BVA	BC-001 (Below)	Executed	**Pass**
TC-BVA-014	Auth Modal (/start)	Password Length	BVA	BC-001 (At)	Executed	**Pass**
TC-BVA-015	Monte Carlo (/monte-carlo)	Horizon (Years)	BVA	BC-007 (Below)	Executed	**Pass**
TC-BVA-016	Monte Carlo (/monte-carlo)	Horizon (Years)	BVA	BC-007 (At)	Executed	**Pass**
TC-BVA-017	Safe Withdrawal (/safe-withdrawal)	Withdrawal Rate (%)	BVA	BC-008 (Below)	Executed	**Pass**
TC-BVA-018	Safe Withdrawal (/safe-withdrawal)	Withdrawal Rate (%)	BVA	BC-008 (At)	Executed	**Pass**
TC-BVA-019	Contact Form (/contact)	Message Length	BVA	BC-009 (Below)	Executed	**Pass**
TC-BVA-020	Contact Form (/contact)	Message Length	BVA	BC-009 (At)	Not Executed	**Pending**
TC-BVA-021	Contact Form (/contact)	Message Length	BVA	BC-010 (At)	Not Executed	**Pending**
TC-BVA-022	Contact Form (/contact)	Message Length	BVA	BC-010 (Above)	Not Executed	**Pending**
TC-BVA-023	Contact Form (/contact)	Subject Length	BVA	BC-011 (At)	Not Executed	**Pending**
TC-BVA-024	Contact Form (/contact)	Subject Length	BVA	BC-011 (Above)	Not Executed	**Pending**
TC-BVA-025	Lump Sum vs DCA (/lumpsum-vs-dca)	Initial Lump Sum ($)	BVA	BC-012 (Below)	Not Executed	**Pending**
TC-BVA-026	Lump Sum vs DCA (/lumpsum-vs-dca)	Initial Lump Sum ($)	BVA	BC-012 (At)	Not Executed	**Pending**
TC-BVA-027	TVM Calculator (/time-value-of-money)	Monthly Contribution ($)	BVA	BC-013 (Below)	Not Executed	**Pending**
TC-BVA-028	TVM Calculator (/time-value-of-money)	Monthly Contribution ($)	BVA	BC-013 (At)	Not Executed	**Pending**
TC-BVA-029	TVM Calculator (/time-value-of-money)	Monthly Contribution ($)	BVA	BC-014 (At)	Not Executed	**Pending**
TC-BVA-030	TVM Calculator (/time-value-of-money)	Monthly Contribution ($)	BVA	BC-014 (Above)	Not Executed	**Pending**
TC-BVA-031	TVM Calculator (/time-value-of-money)	Starting Value ($)	BVA	BC-015 (Below)	Not Executed	**Pending**
TC-BVA-032	TVM Calculator (/time-value-of-money)	Starting Value ($)	BVA	BC-015 (At)	Not Executed	**Pending**

D5 — Execution Log [20 marks]

ℹ *For every EXECUTED test case: cite its TC ID and paste screenshot(s) proving the test was run. A single screenshot capturing both the input and the system's response is fine; otherwise use two. Verdict, actual result, and defect notes belong in your D2/D3 tables — no need to repeat them here.*

Executed Test Case 1

TC ID	TC-EP-003

Screenshot(s) — paste below (input + response; one combined screenshot is acceptable):

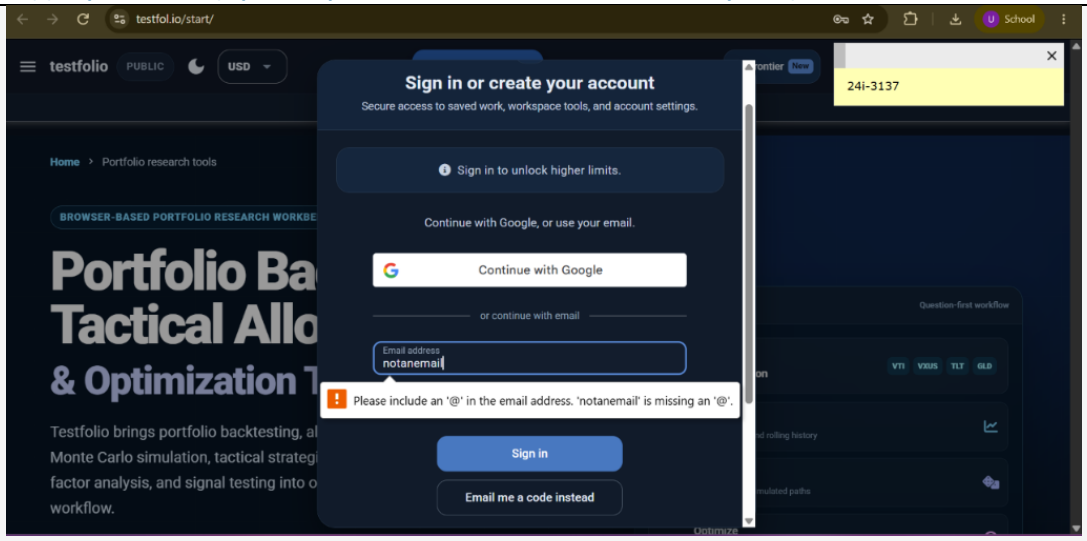


Figure E-01: Auth Modal (/start): Malformed email missing '@' trapped by native HTML5 validation tooltip. (Student Roll No: 24i-3137)

Executed Test Case 2

TC ID	TC-EP-004

Screenshot(s) — paste below (input + response; one combined screenshot is acceptable):

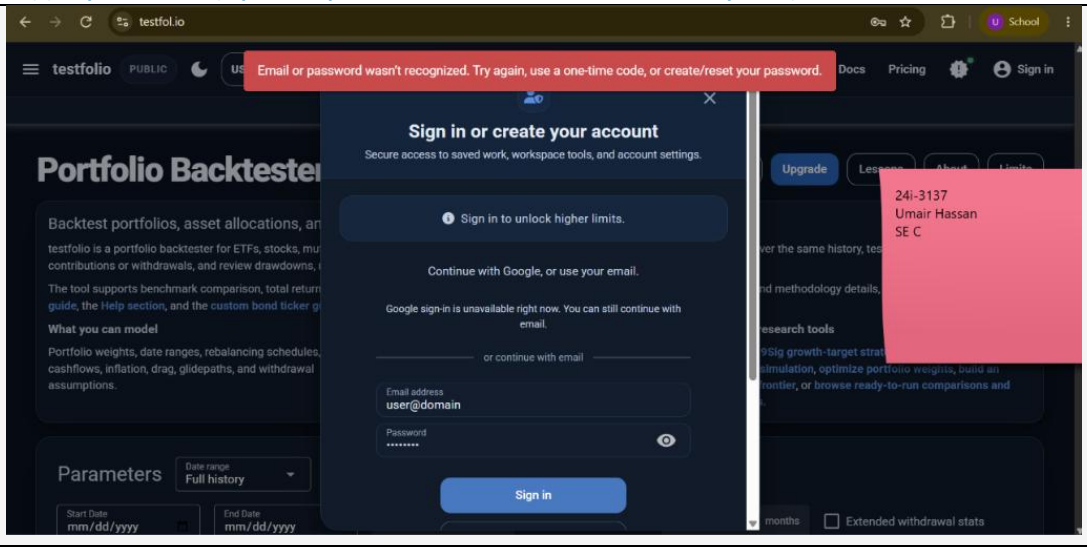


Figure E-02: Auth Modal (/start): Email missing domain TLD (user@domain) triggers server error banner. (Student Roll No: 24i-3137)

Executed Test Case 3

TC ID	TC-EP-005

Screenshot(s) — paste below (input + response; one combined screenshot is acceptable):

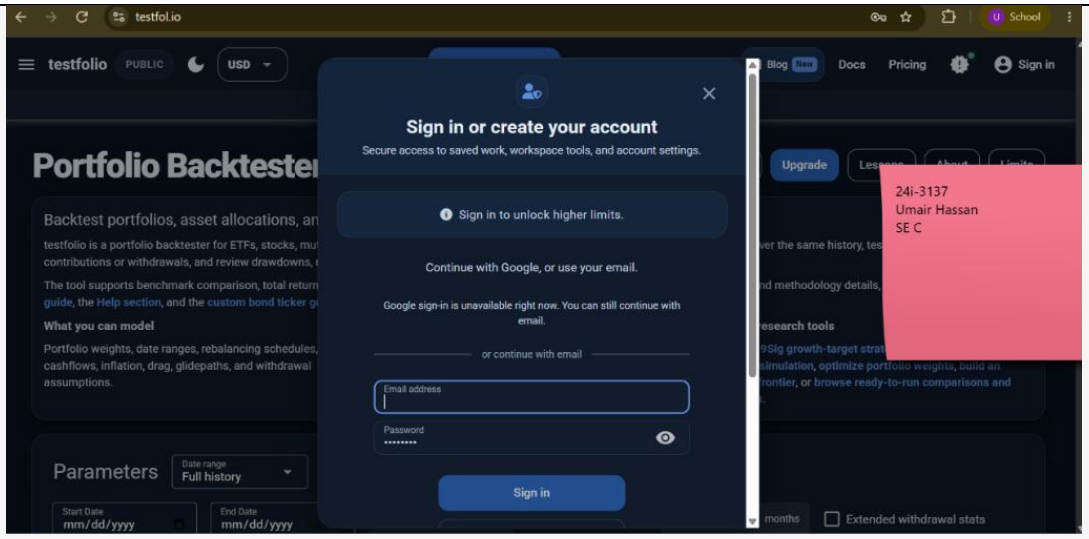


Figure E-03: Auth Modal (/start): Empty email disables submit button client-side. (Student Roll No: 24i-3137)

Executed Test Case 4

TC ID	TC-EP-006

Screenshot(s) — paste below (input + response; one combined screenshot is acceptable):

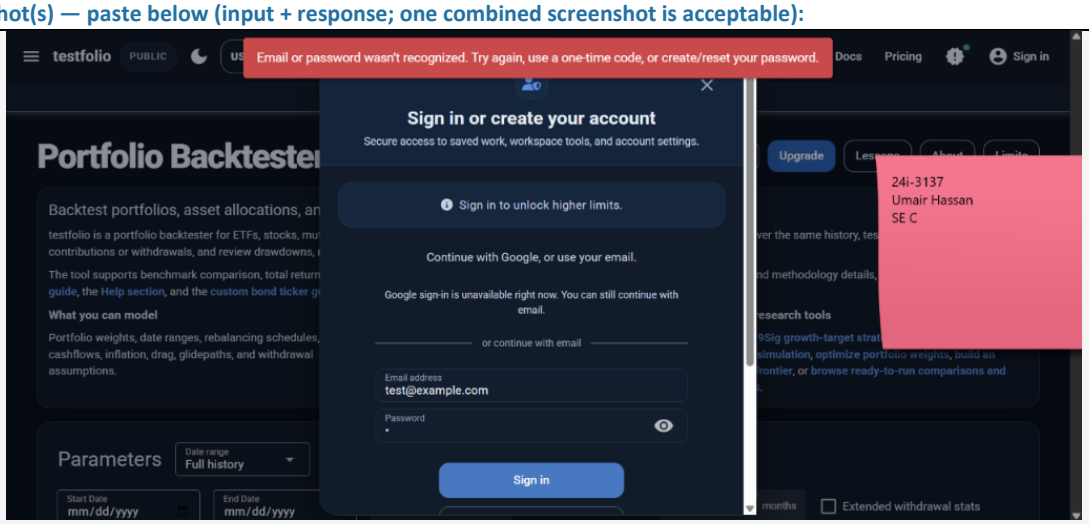


Figure E-04: Auth Modal (/login): 1-char password activates submit button client-side. (Student Roll No: 24i-3137)

Executed Test Case 5

TC ID	TC-EP-007

Screenshot(s) — paste below (input + response; one combined screenshot is acceptable):

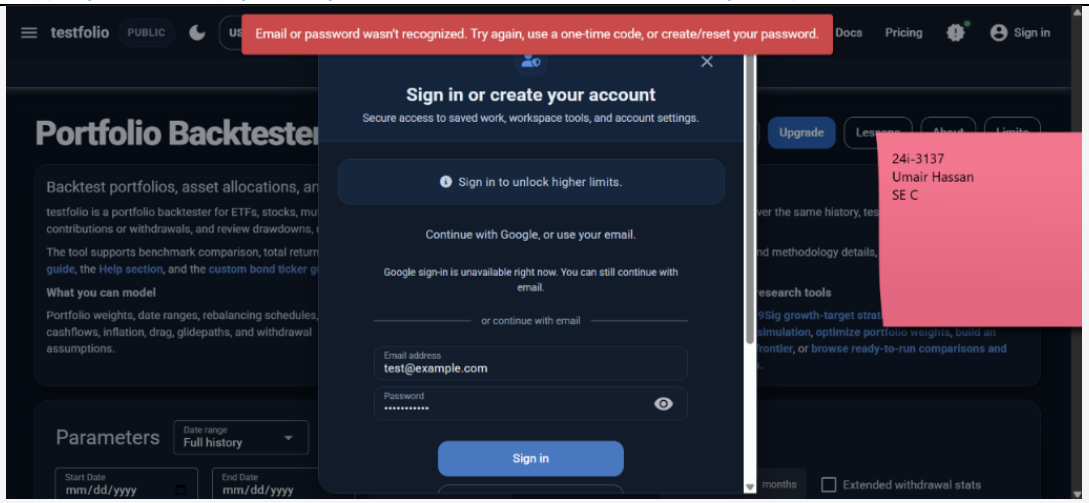


Figure E-05: Auth Modal (/login): Password with special characters accepted by input mask. (Student Roll No: 24i-3137)

↑ *Duplicate the block above for each additional executed test case* ↑

— End of Assignment 3 Template —

TC ID **TC-EP-008**

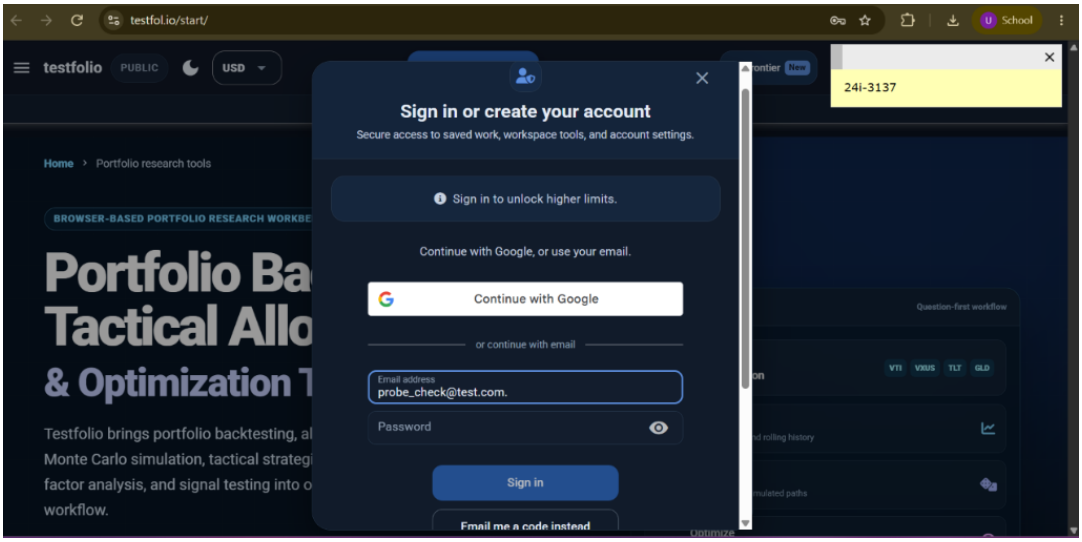


Figure E-06: Auth Modal (/start): Empty Password client-side validation disables submit button. (Student Roll No: 24i-3137)

TC ID **TC-EP-009**

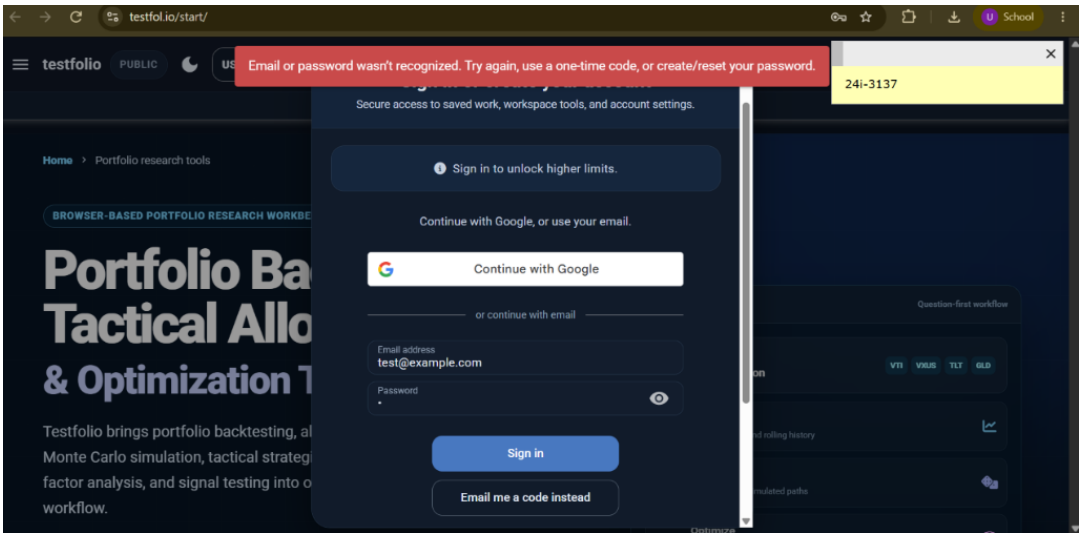


Figure E-07: Auth Modal (/login): Unrecognized credentials error banner displayed. (Student Roll No: 24i-3137)

TC ID **TC-EP-010**

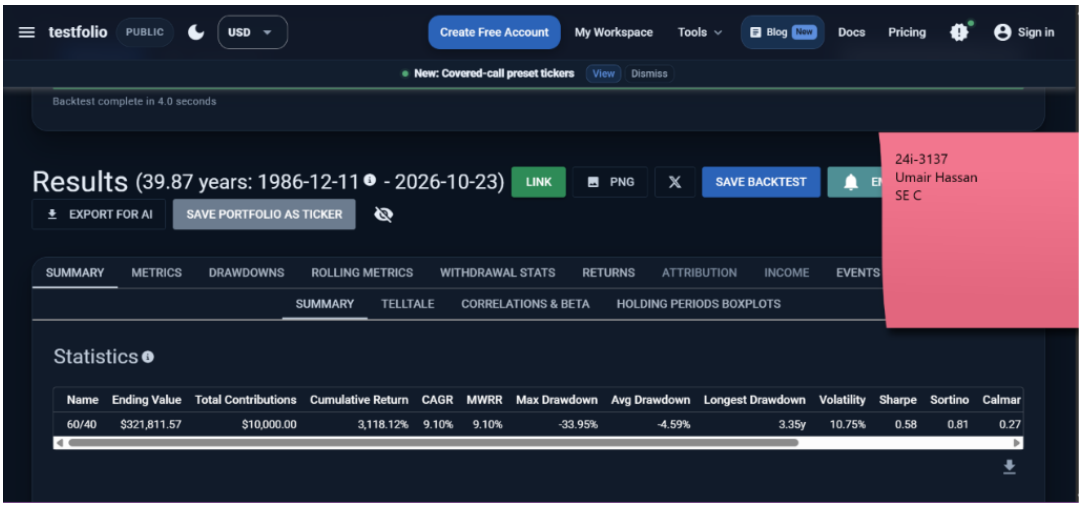


Figure E-08: Portfolio Backtester (/): Valid backtest execution (SPY, CAGR 9.10%, Sharpe 0.58). (Student Roll No: 24i-3137)

TC ID **TC-EP-011**



Figure E-09: Portfolio Backtester (/): Dual portfolio comparative performance (60/40 vs HFEA). (Student Roll No: 24i-3137)

TC ID **TC-EP-012**

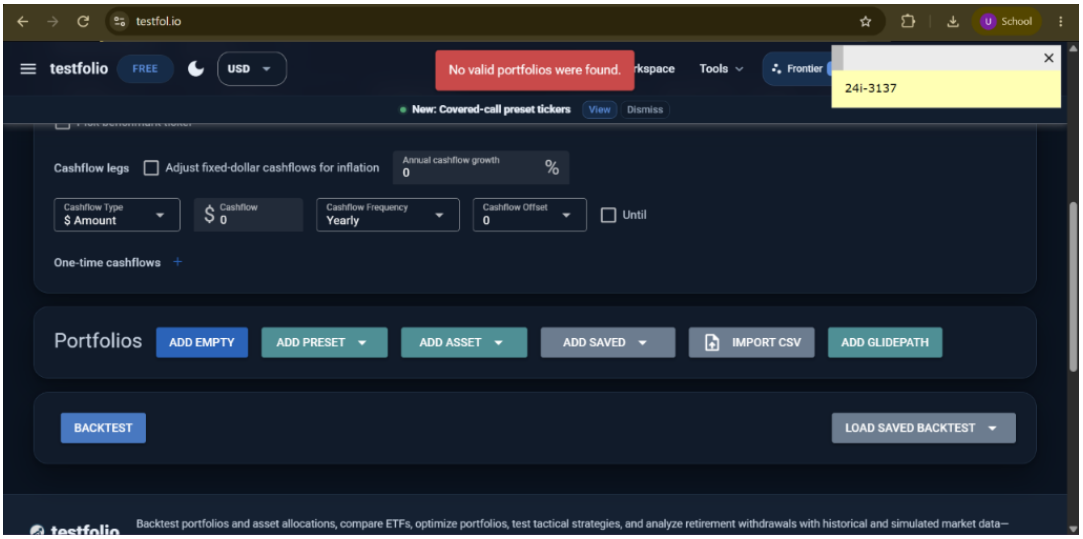


Figure E-10: Portfolio Backtester (/): 0 portfolios added triggers red banner 'No valid portfolios were found.' (Student Roll No: 24i-3137)

TC ID **TC-EP-013**

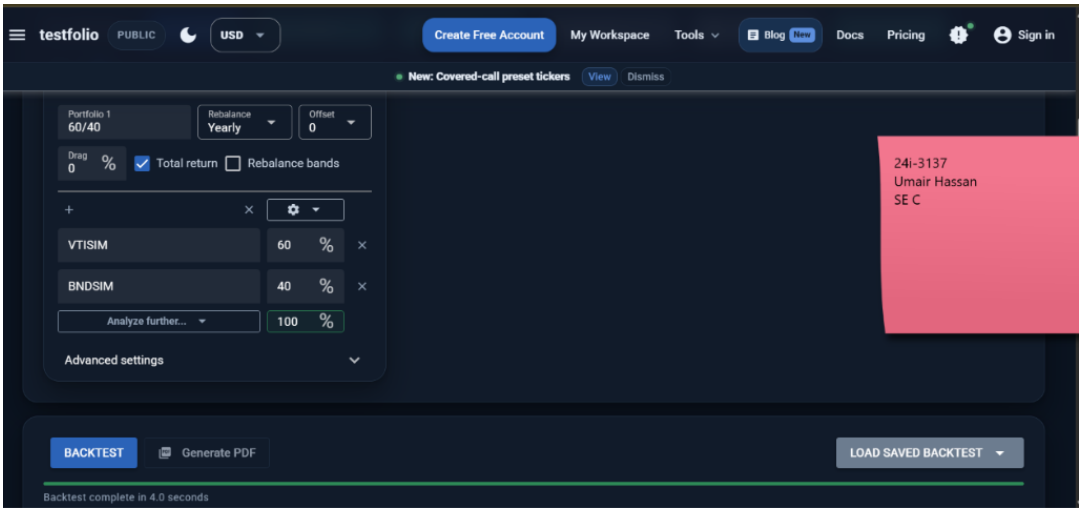


Figure E-11: Portfolio Backtester (/): Exact 100% asset allocation accepted. (Student Roll No: 24i-3137)

TC ID **TC-EP-014**

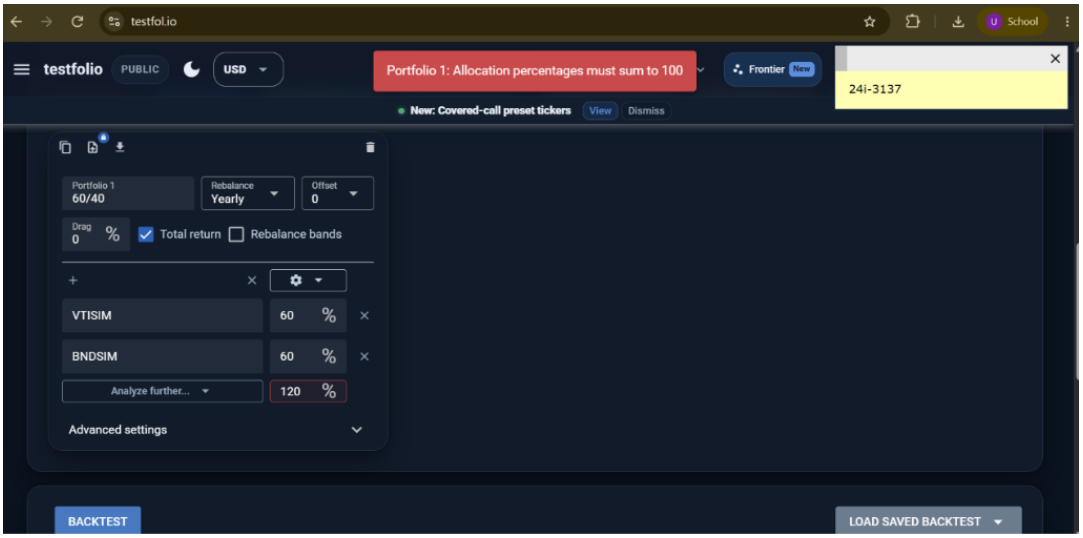


Figure E-12: Portfolio Backtester (/): Allocations summing to 120% (> 100%) triggers red banner 'Allocations must sum to 100'. (Student Roll No: 24i-3137)

TC ID **TC-EP-015**

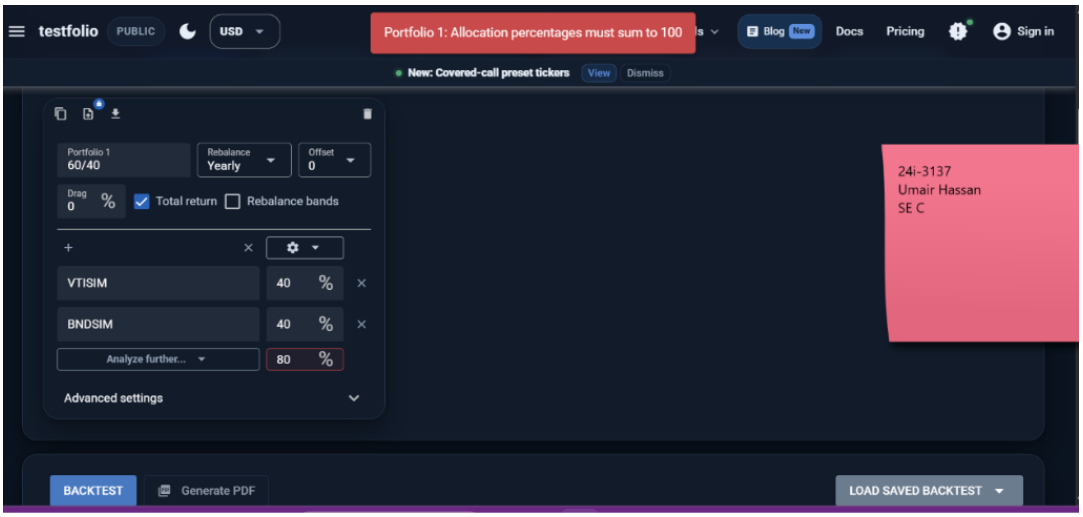


Figure E-13: Portfolio Backtester (/): Under-allocation 80% triggers banner 'Allocations must sum to 100'. (Student Roll No: 24i-3137)

TC ID **TC-EP-016**

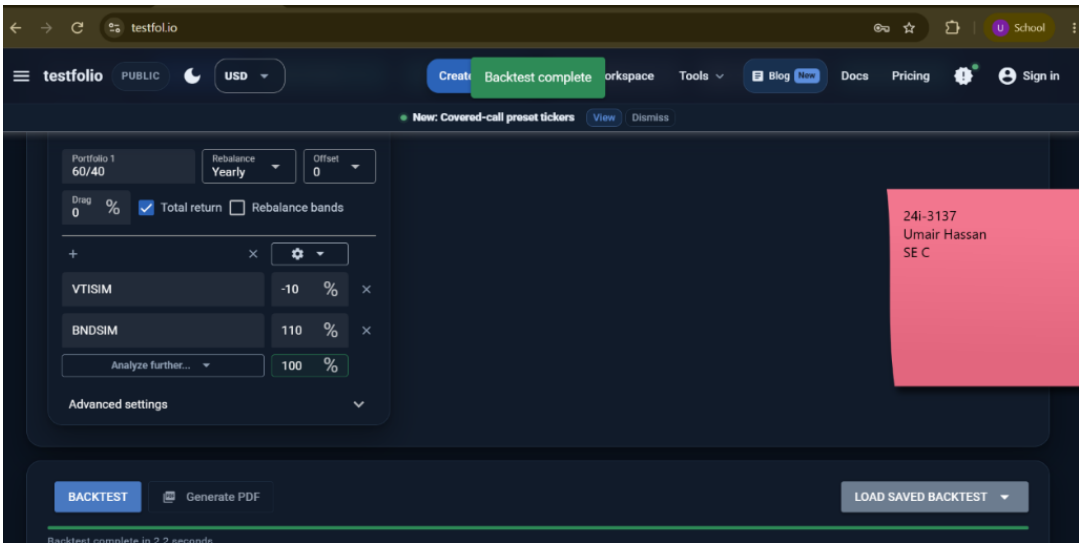


Figure E-14: Portfolio Backtester (/): Negative allocation -10% (shorting accepted, total sums to 100%). (Student Roll No: 24i-3137)

TC ID **TC-EP-020**

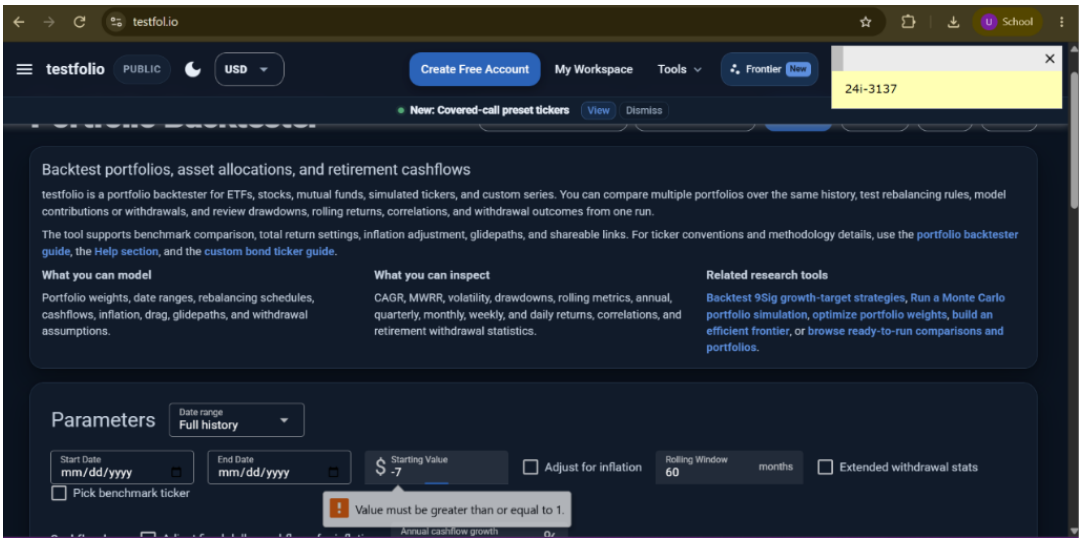


Figure E-15: Portfolio Backtester (/): Starting Value = -$7 triggers tooltip 'Value must be greater than or equal to 1.' (Student Roll No: 24i-3137)

TC ID **TC-EP-021**

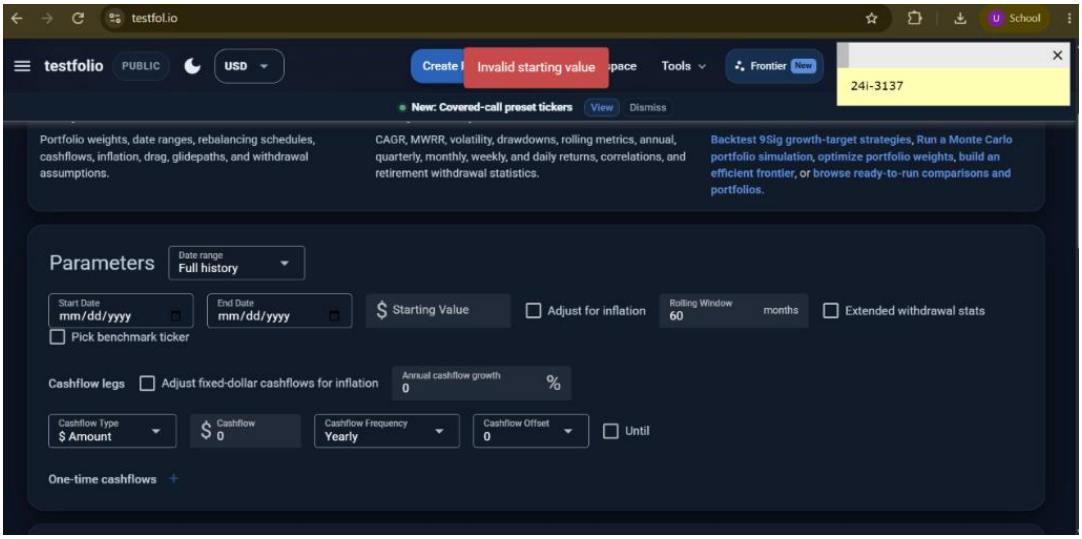


Figure E-16: Portfolio Backtester (/): Empty starting value triggers application red banner 'Invalid starting value'. (Student Roll No: 24i-3137)

TC ID **TC-EP-023**

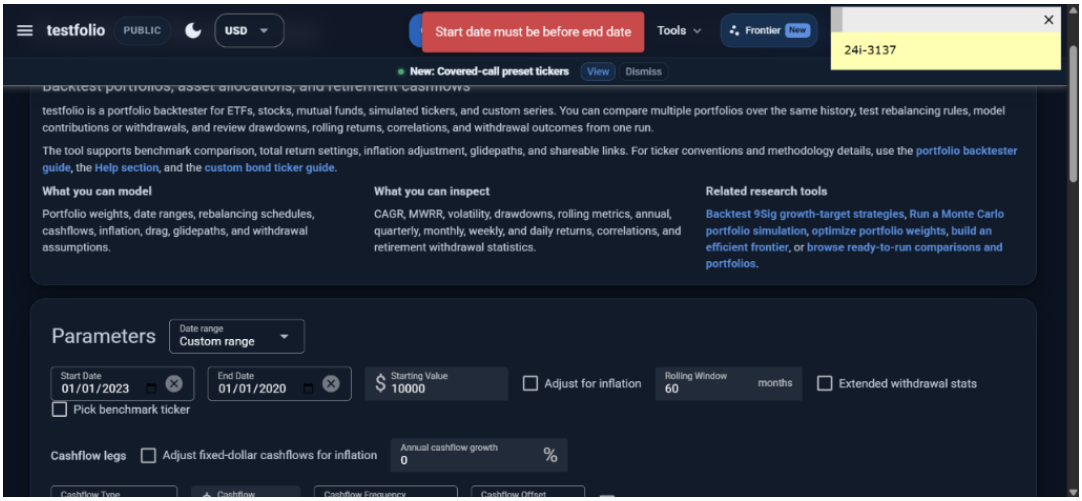


Figure E-17: Portfolio Backtester (/): Inverted date range (Start 2023 > End 2020) triggers 'Start date must be before end date'. (Student Roll No: 24i-3137)

TC ID **TC-EP-024**

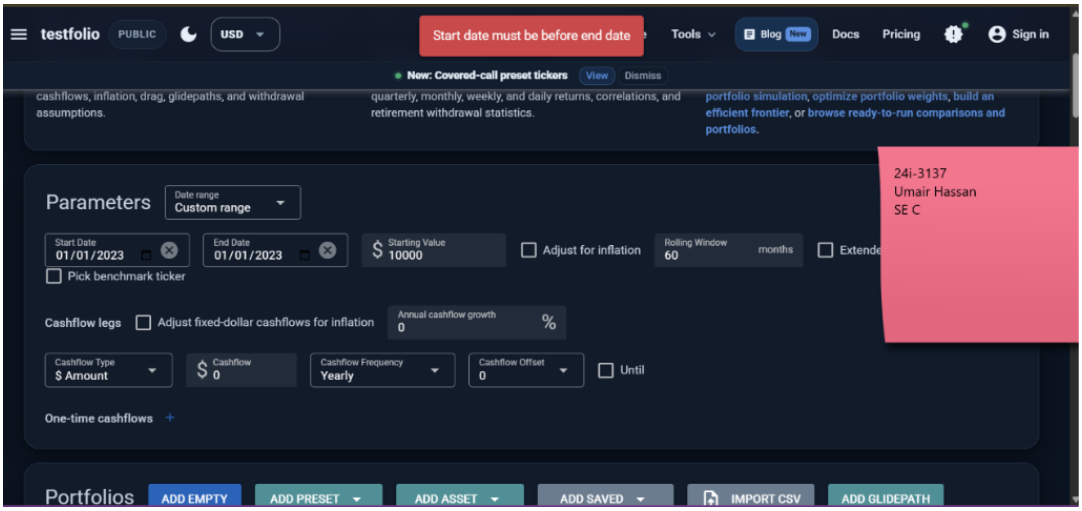


Figure E-18: Portfolio Backtester (/): Start date == End date triggers 'Start date must be before end date'. (Student Roll No: 24i-3137)

TC ID **TC-EP-025**

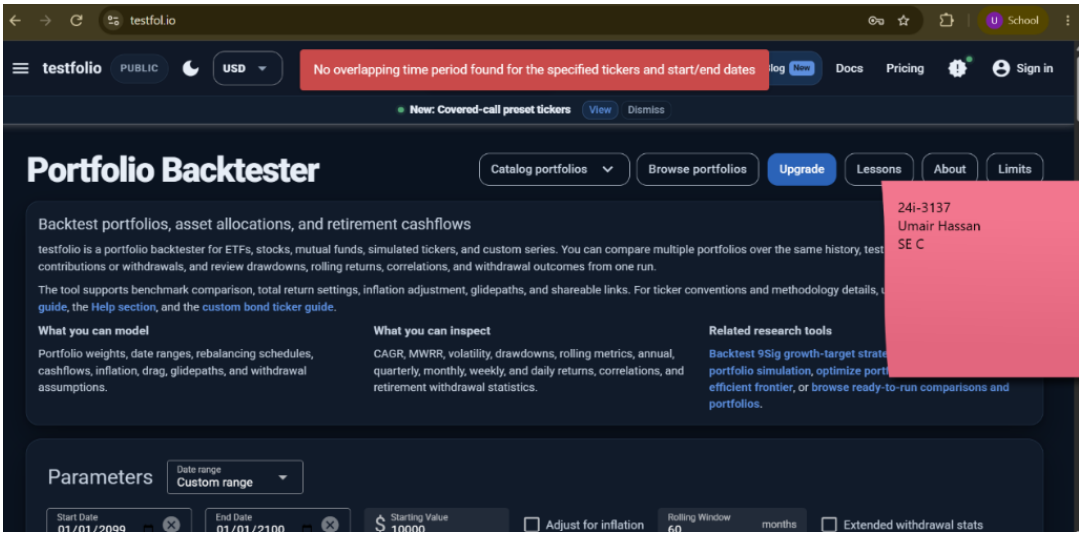


Figure E-19: Portfolio Backtester (/): Future date range (2099-2100) triggers 'No overlapping time period found'. (Student Roll No: 24i-3137)

TC ID **TC-EP-027**

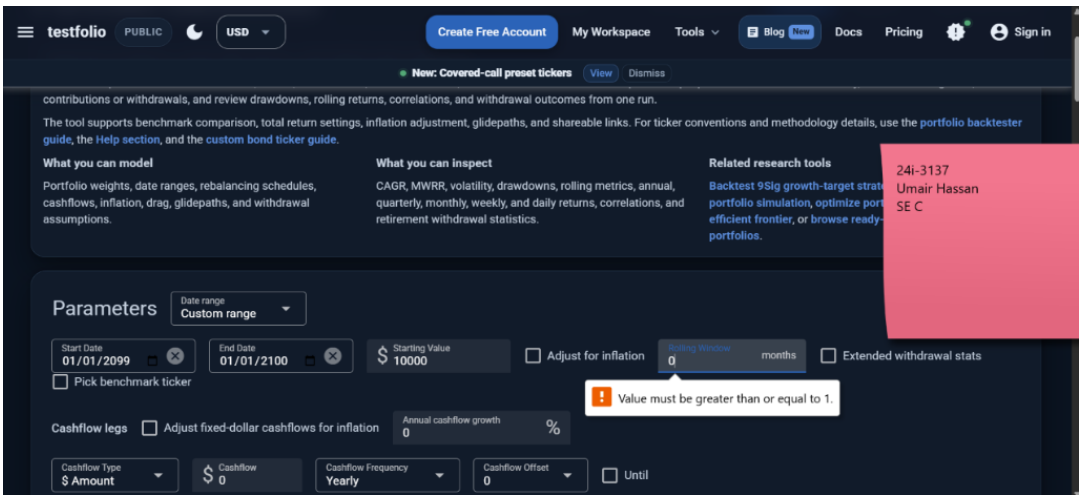


Figure E-20: Portfolio Backtester (/): Rolling window 0 triggers tooltip 'Value must be greater than or equal to 1.' (Student Roll No: 24i-3137)

TC ID **TC-EP-028**

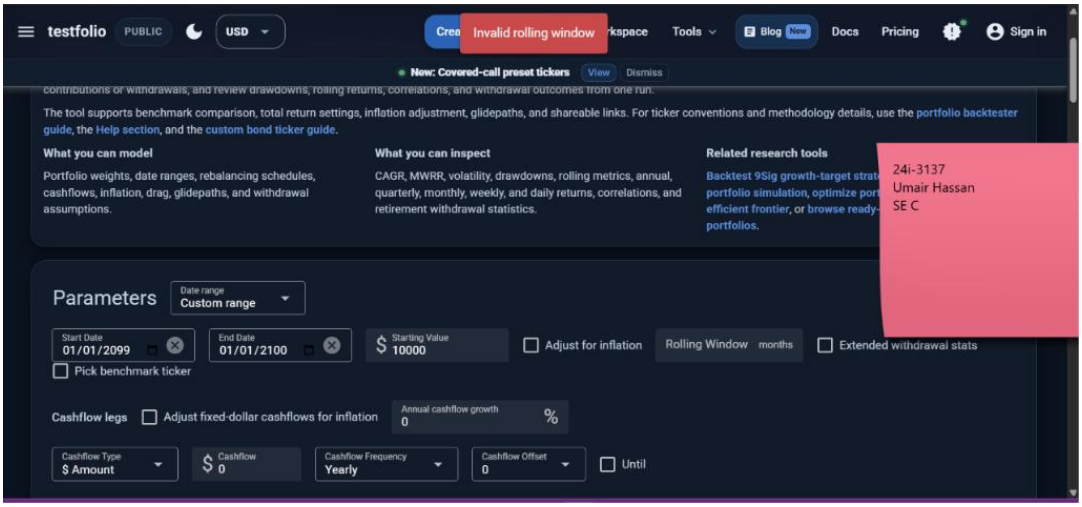


Figure E-21: Portfolio Backtester (/): Empty rolling window triggers red banner 'Invalid rolling window'. (Student Roll No: 24i-3137)

TC ID **TC-EP-030**

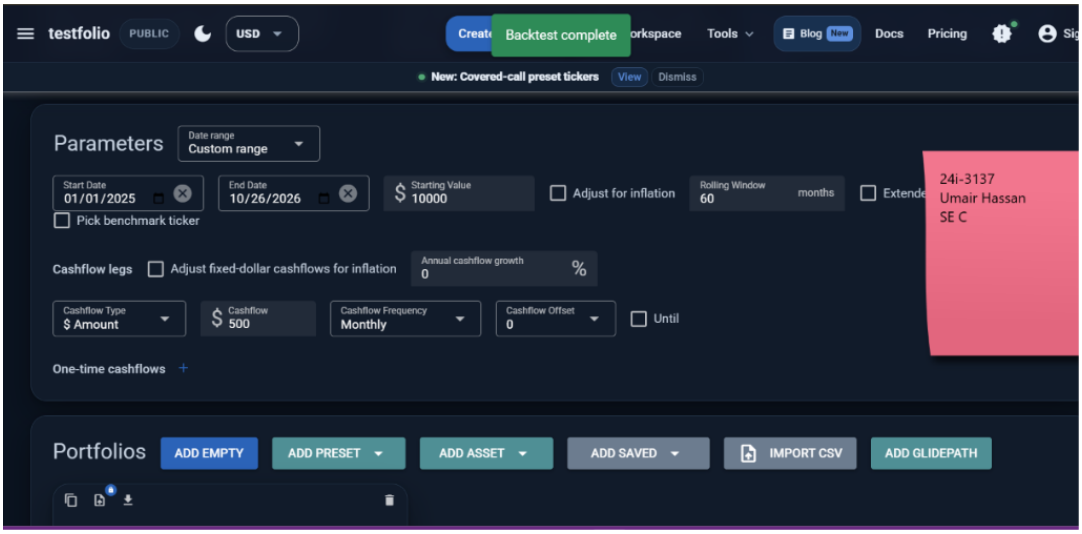


Figure E-22: Portfolio Backtester (/): Monthly cashflow $500 backtest completed successfully. (Student Roll No: 24i-3137)

TC ID **TC-EP-031**

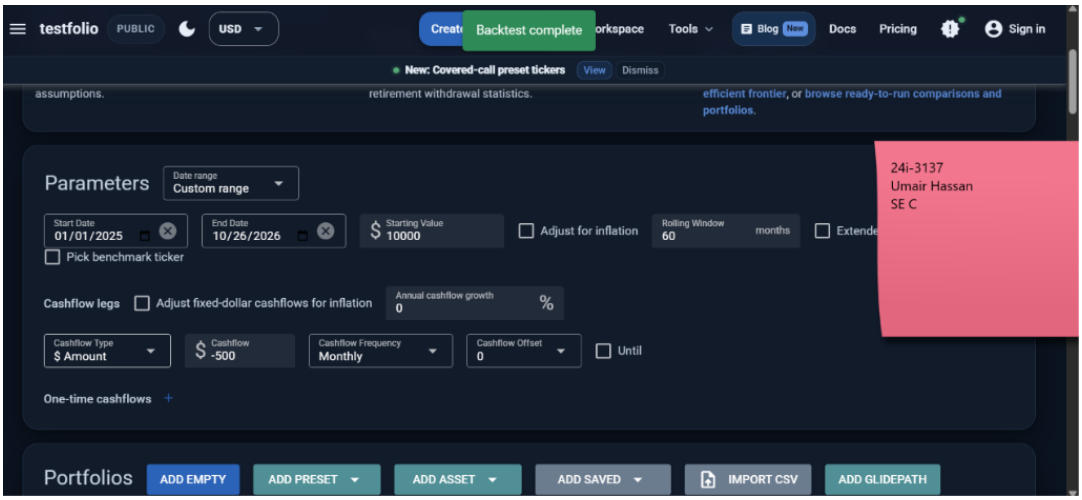


Figure E-23: Portfolio Backtester (/): Monthly negative cashflow -$500 accepted and executed as withdrawal leg. (Student Roll No: 24i-3137)

TC ID **TC-EP-032**

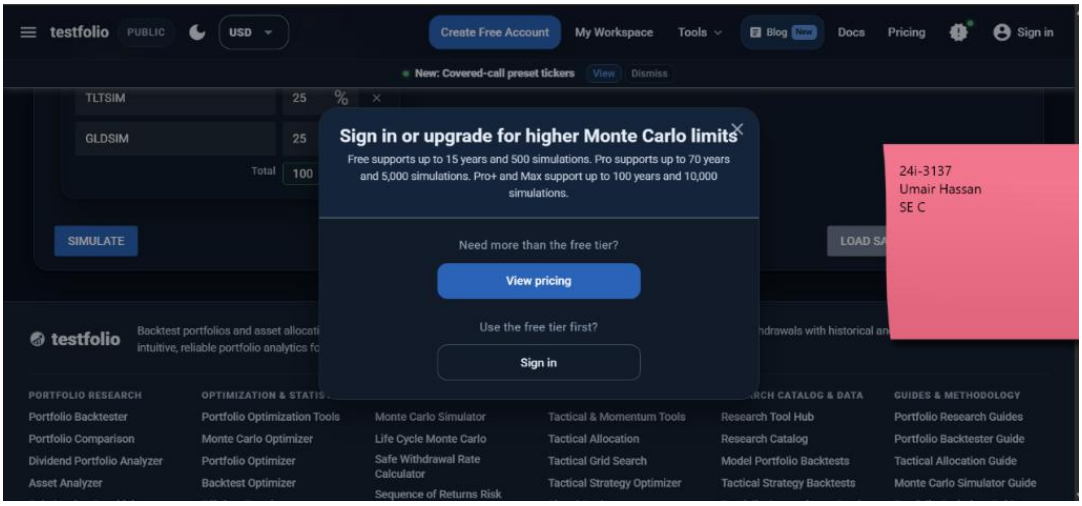


Figure E-24: Monte Carlo Simulator (/monte-carlo): Horizon 30 yrs triggers modal 'Sign in or upgrade for higher Monte Carlo limits'. (Student Roll No: 24i-3137)

TC ID **TC-EP-033**

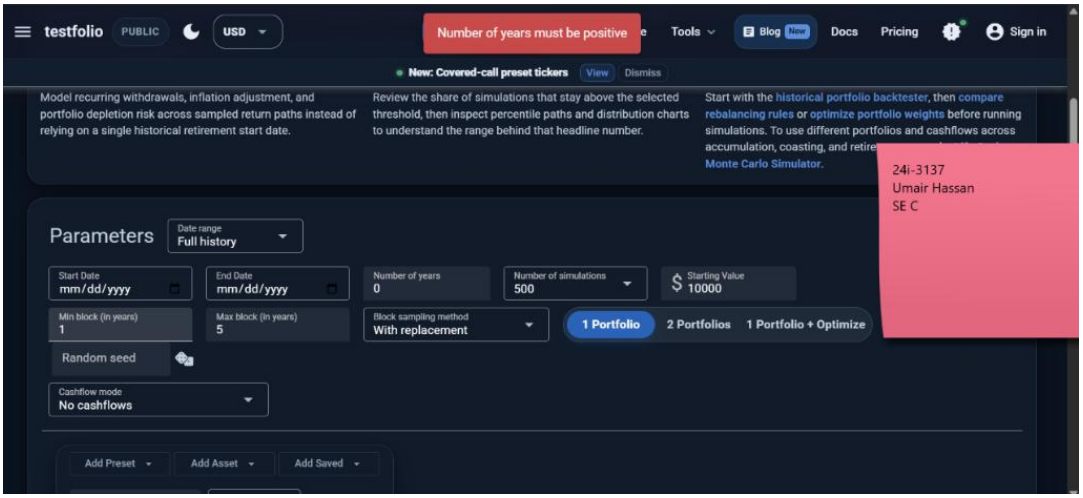


Figure E-25: Monte Carlo Simulator (/monte-carlo): Horizon 0 yrs triggers banner 'Number of years must be positive'. (Student Roll No: 24i-3137)

TC ID **TC-EP-034**

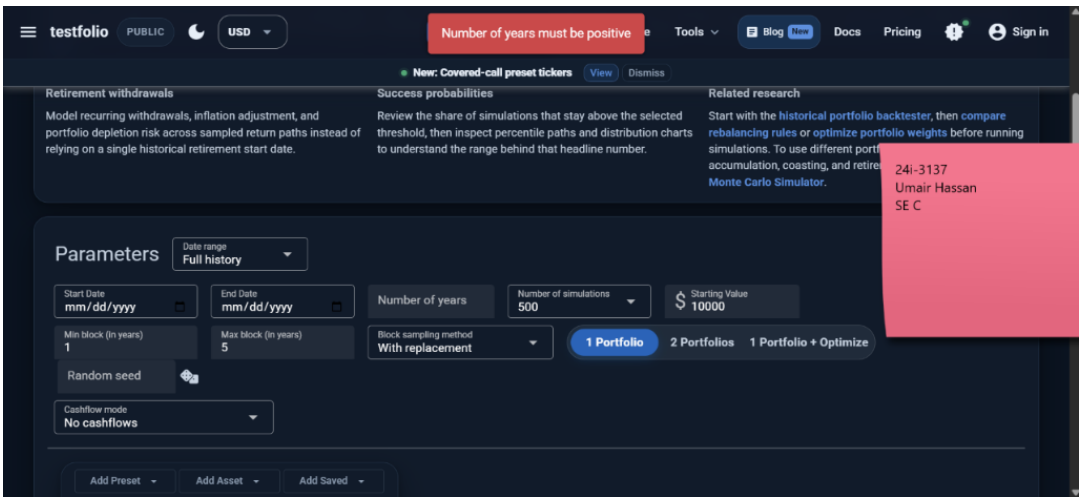


Figure E-26: Monte Carlo Simulator (/monte-carlo): Empty horizon triggers banner 'Number of years must be positive'. (Student Roll No: 24i-3137)

TC ID **TC-EP-037**

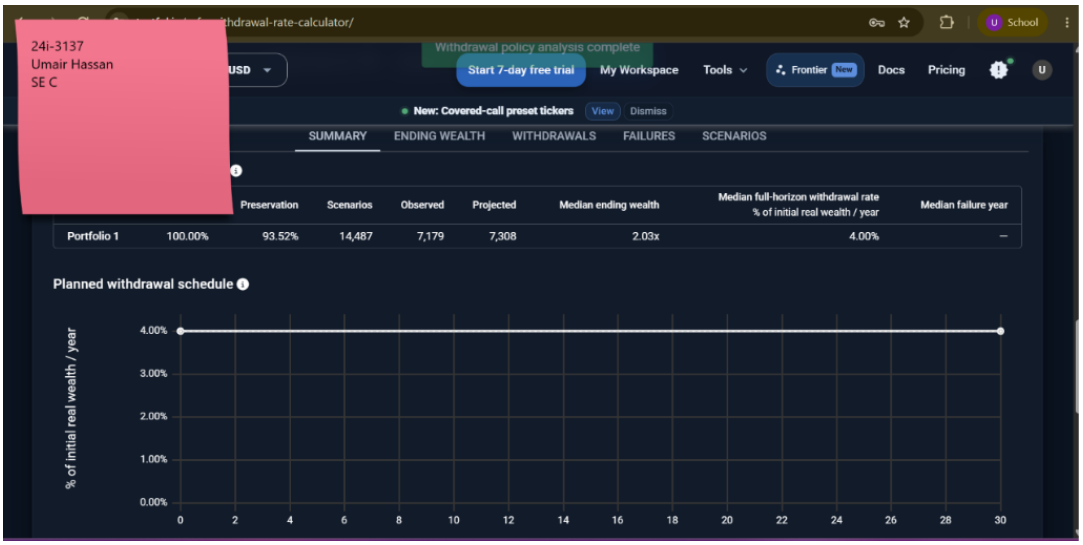


Figure E-27: Safe Withdrawal Rate (/safe-withdrawal-rate-calculator): 4.00% baseline executed (93.52% preservation). (Student Roll No: 24i-3137)

TC ID **TC-EP-038**

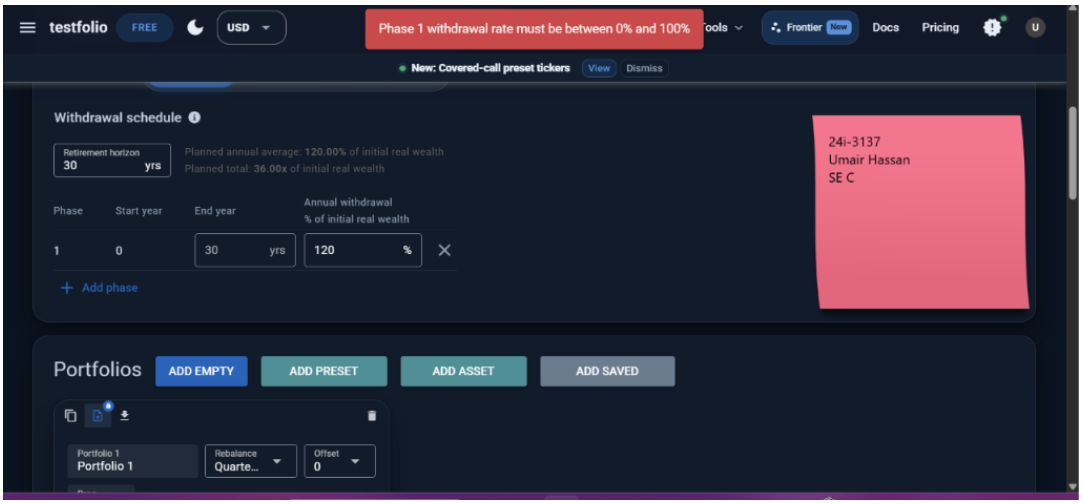


Figure E-28: Safe Withdrawal Rate (/safe-withdrawal-rate-calculator): 120% rate triggers banner 'Phase 1 withdrawal rate must be between 0% and 100%'. (Student Roll No: 24i-3137)

TC ID **TC-EP-039**

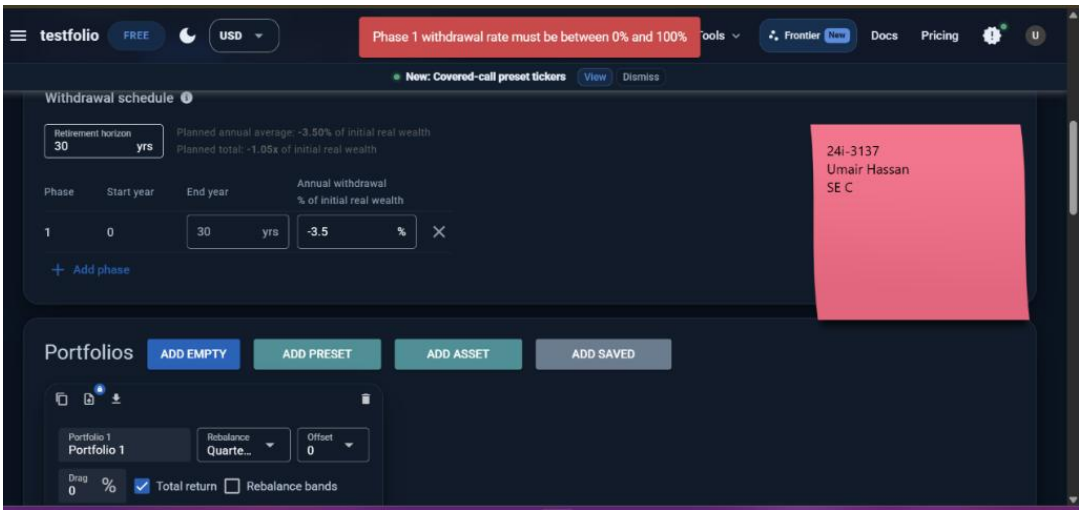


Figure E-29: Safe Withdrawal Rate (/safe-withdrawal-rate-calculator): Negative -3.5% rate triggers banner 'Phase 1 withdrawal rate must be between 0% and 100%'. (Student Roll No: 24i-3137)

TC ID **TC-EP-040**

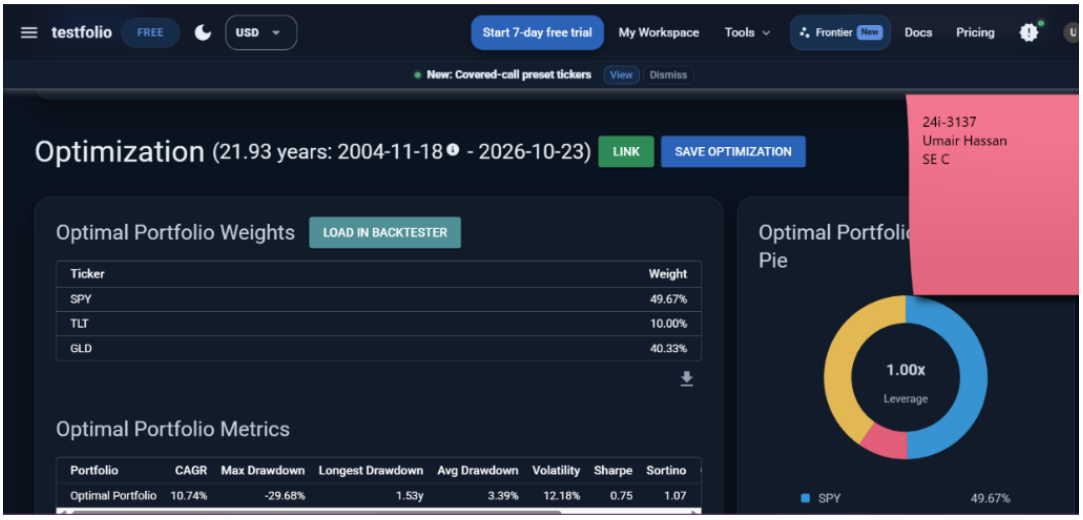


Figure E-30: Backtest Optimizer (/backtest-optimizer): Bounded optimization solved (SPY 49.67%, TLT 10%, GLD 40.33%). (Student Roll No: 24i-3137)

TC ID **TC-EP-041**

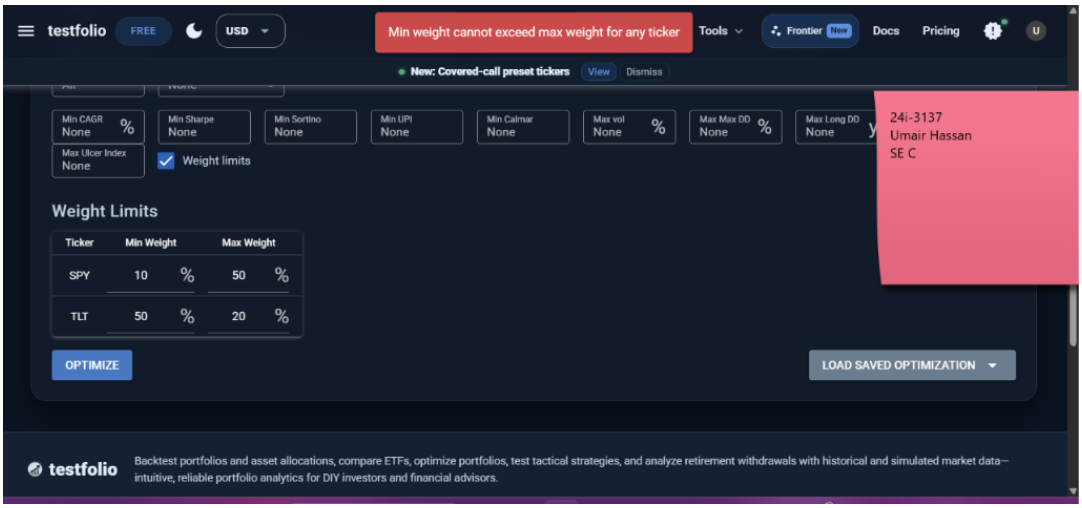


Figure E-31: Backtest Optimizer (/backtest-optimizer): Inconsistent bounds Min > Max triggers 'Min weight cannot exceed max weight for any ticker'. (Student Roll No: 24i-3137)

TC ID **TC-EP-042**

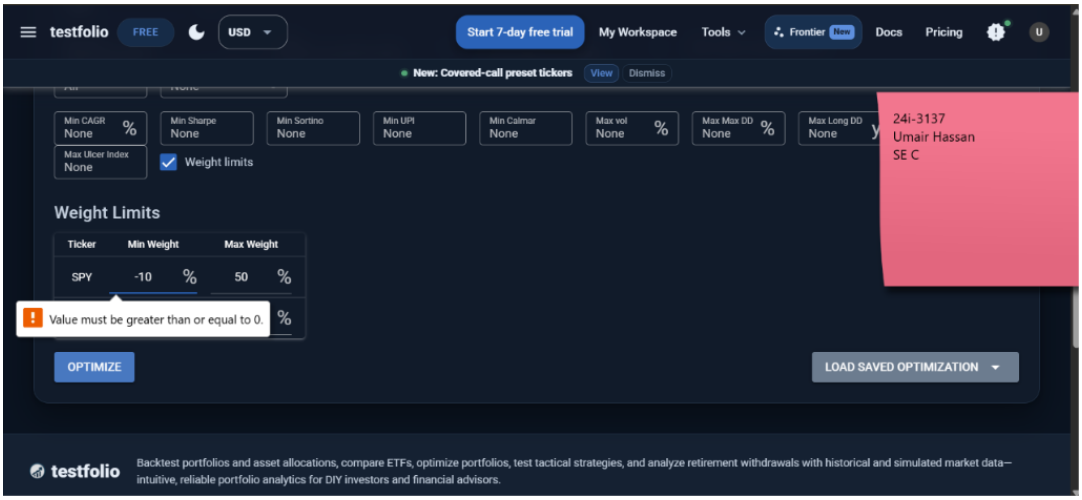


Figure E-32: Backtest Optimizer (/backtest-optimizer): Negative bound -10% triggers tooltip 'Value must be greater than or equal to 0.' (Student Roll No: 24i-3137)

TC ID **TC-EP-043**



Figure E-33: Efficient Frontier (/efficient-frontier): Multi-asset covariance curved efficient frontier rendered. (Student Roll No: 24i-3137)

TC ID **TC-EP-044**

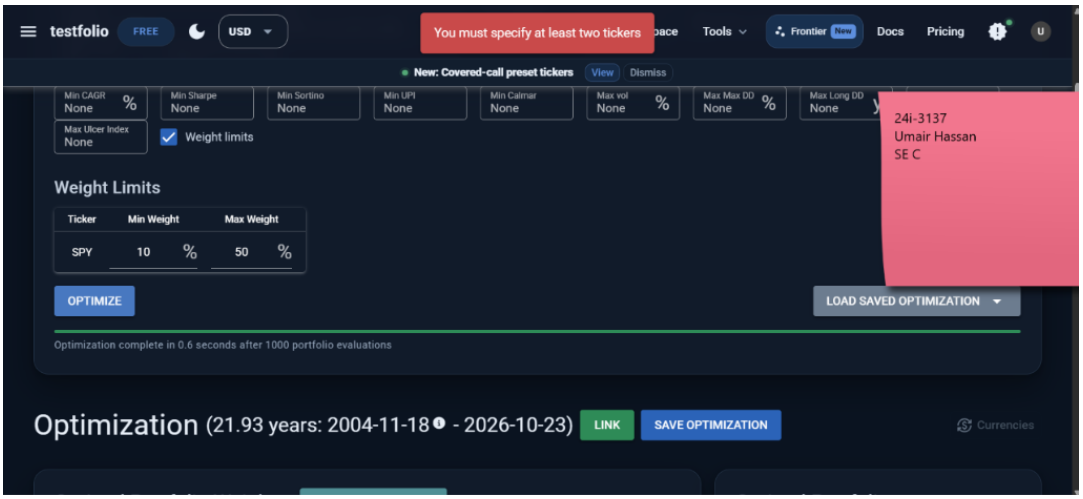


Figure E-34: Backtest Optimizer / Frontier: Single ticker SPY triggers banner 'You must specify at least two tickers'. (Student Roll No: 24i-3137)

TC ID **TC-EP-052**

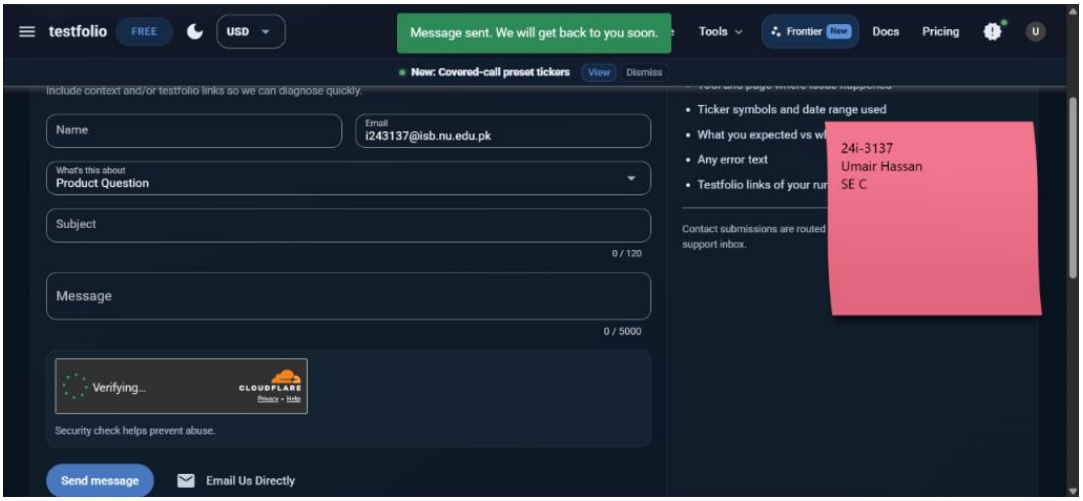


Figure E-35: Contact Form (/contact): Valid inquiry submitted with student email i243137@isb.nu.edu.pk; green banner 'Message sent'. (Student Roll No: 24i-3137)

TC ID **TC-EP-053**

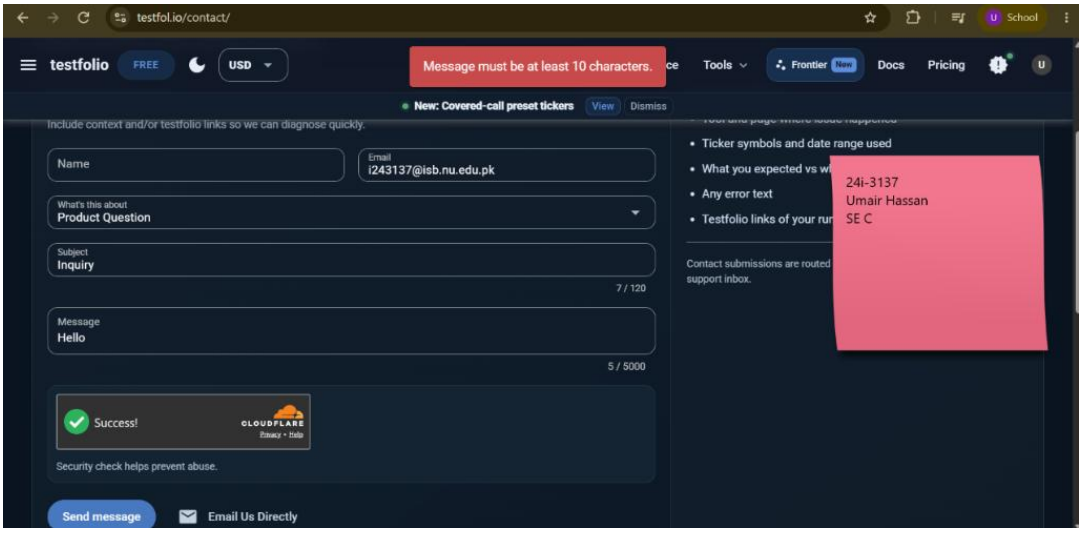


Figure E-36: Contact Form (/contact): Message length = 5 chars triggers red banner 'Message must be at least 10 characters.' (Student Roll No: 24i-3137)

TC ID **TC-EP-057**

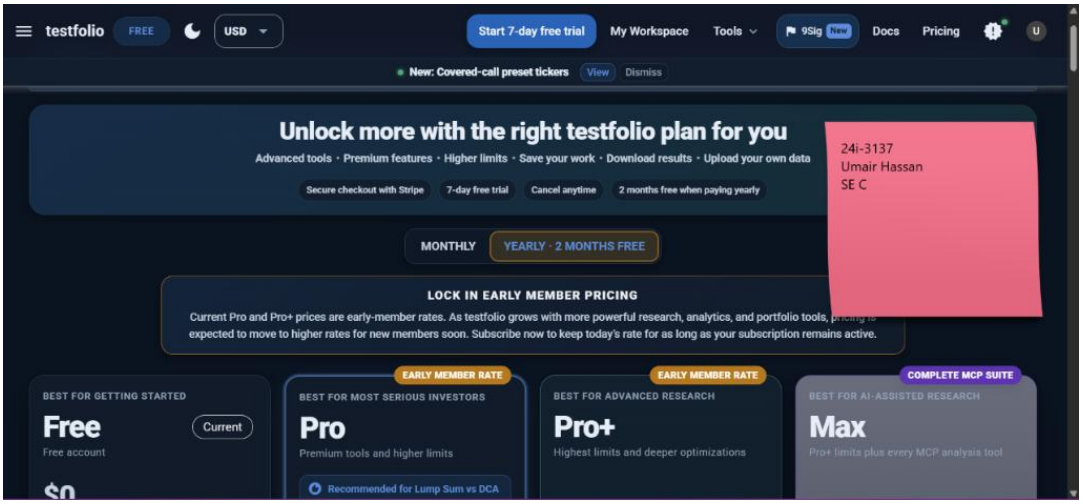


Figure E-37: Lump Sum vs DCA (/lumpsum-vs-dca): Feature gated by paywall; redirects to pricing table 'Recommended for Lump Sum vs DCA'. (Student Roll No: 24i-3137)

TC ID **TC-EP-062**

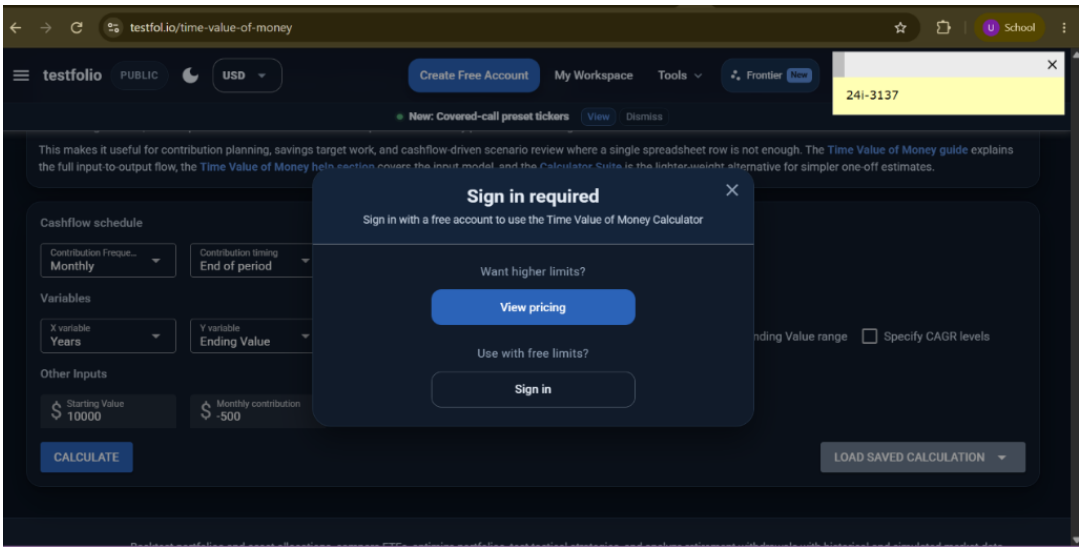


Figure E-38: Time Value of Money (/time-value-of-money): Gated calculation triggers 'Sign in required' modal dialog. (Student Roll No: 24i-3137)

TC ID **TC-EP-065**

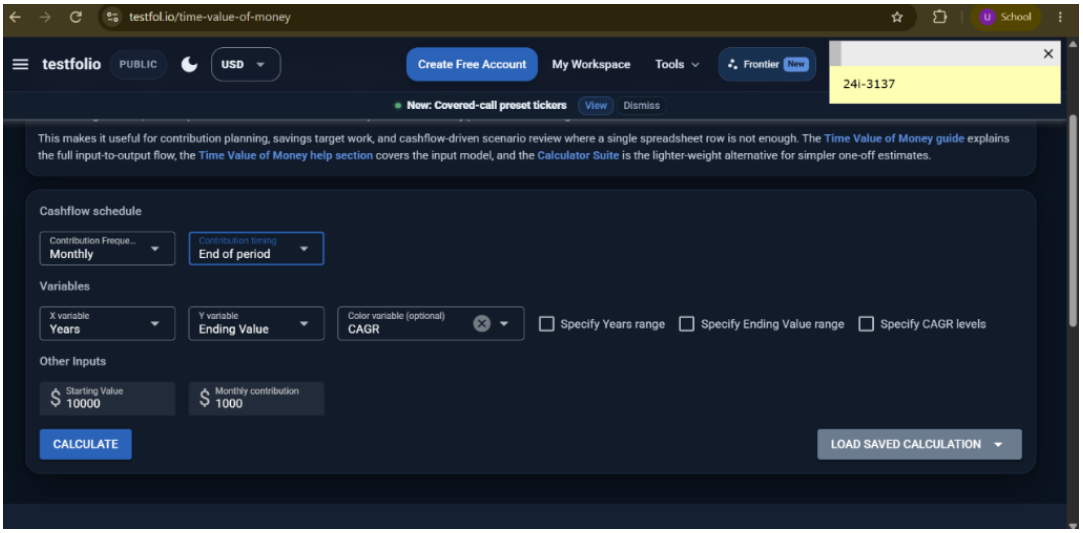


Figure E-39: Time Value of Money (/time-value-of-money): Form inputs ($10,000 starting value, $1,000 monthly contribution). (Student Roll No: 24i-3137)

TC ID **TC-EP-066**

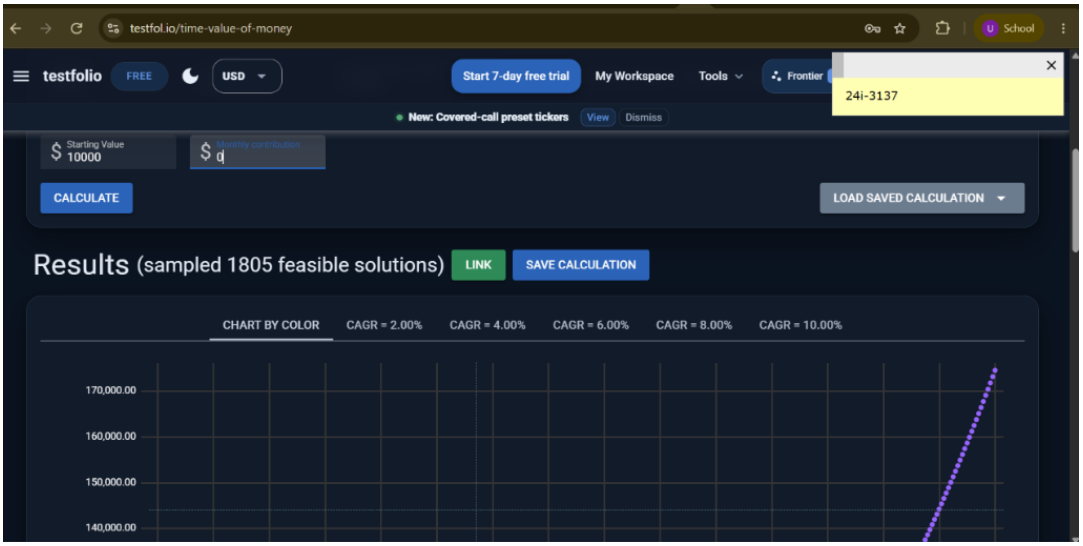


Figure E-40: Time Value of Money (/time-value-of-money): Monthly contribution = $0 boundary computes 1,805 solutions. (Student Roll No: 24i-3137)

TC ID **TC-EP-067**

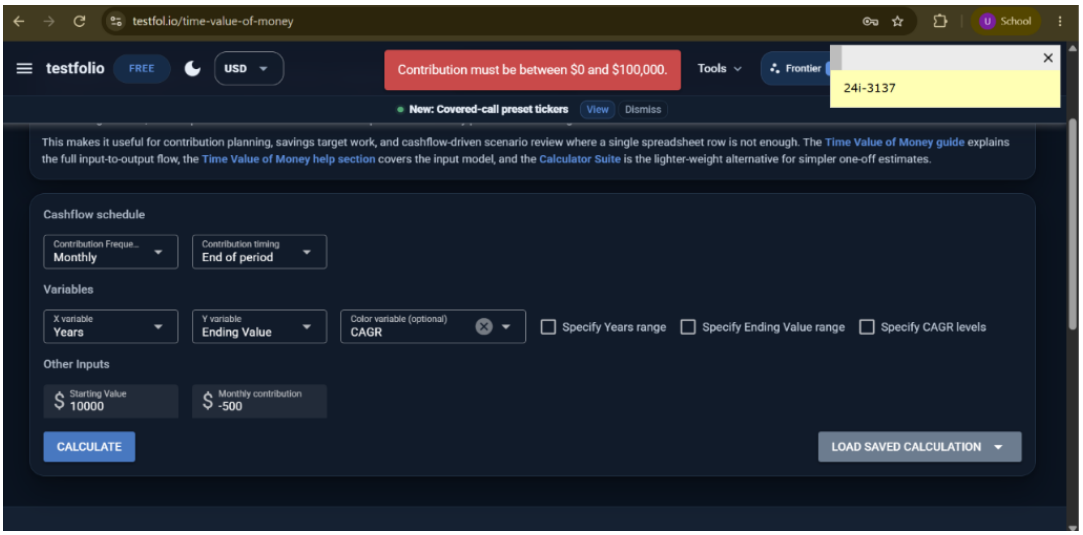


Figure E-41: Time Value of Money (/time-value-of-money): Monthly contribution = -$500 triggers 'Contribution must be between $0 and $100,000.' (Student Roll No: 24i-3137)

TC ID **TC-EP-069**

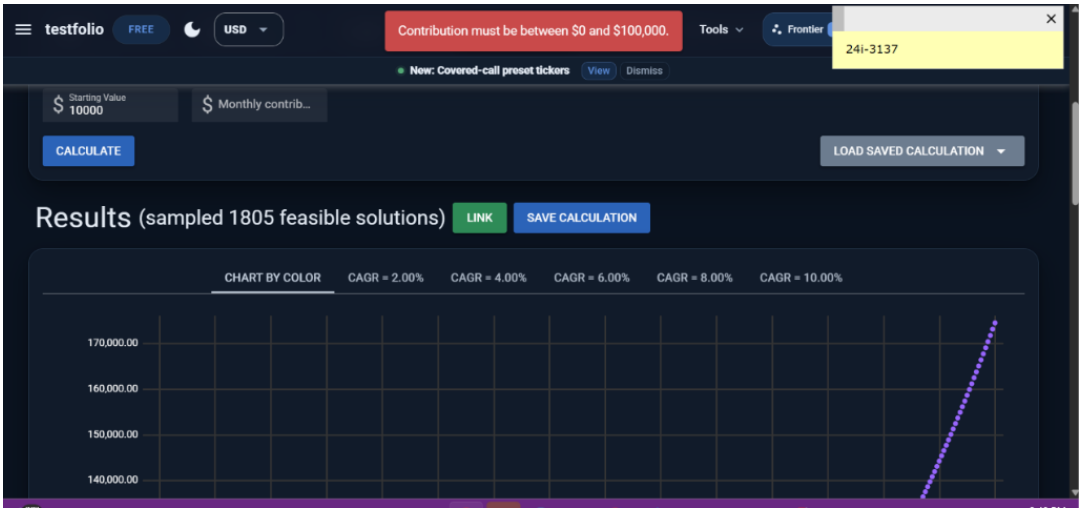


Figure E-42: Time Value of Money (/time-value-of-money): Empty contribution triggers error banner while retaining prior graph (Defect). (Student Roll No: 24i-3137)

TC ID **TC-EP-070**

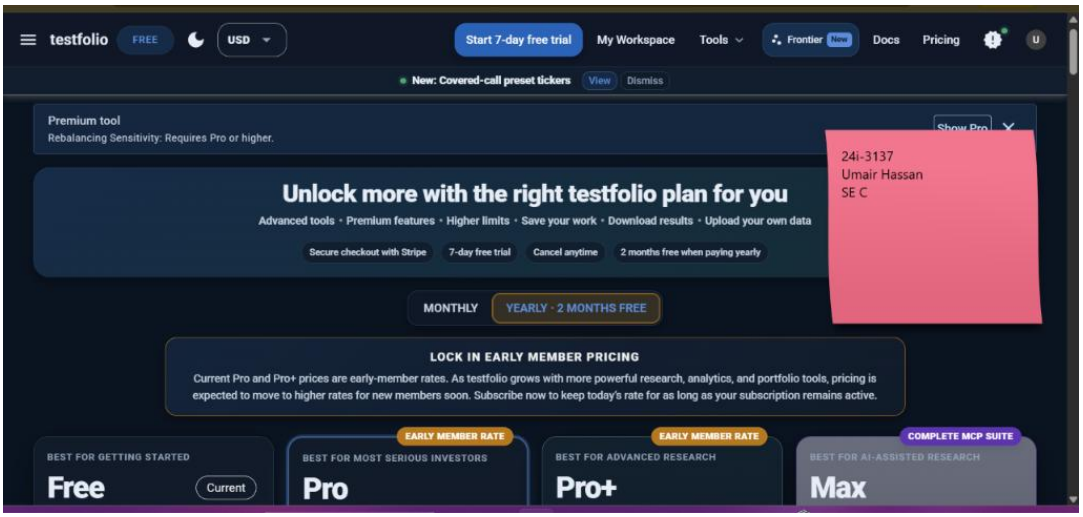


Figure E-43: Rebalancing Sensitivity (/rebalancing-sensitivity): Gated premium tool; banner 'Premium tool: Requires Pro or higher.' (Student Roll No: 24i-3137)

TC ID **TC-BVA-001**

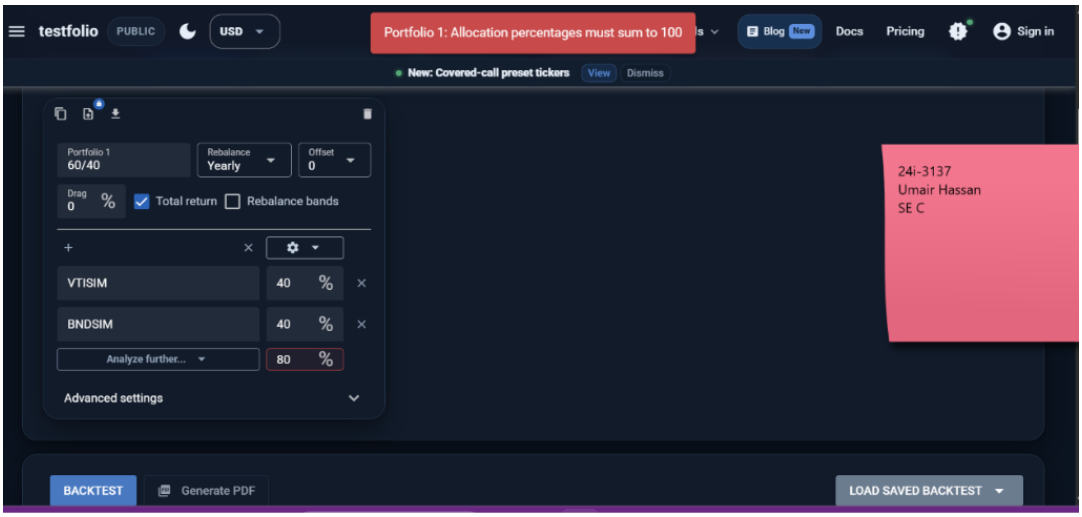


Figure E-44: Portfolio Backtester (/): BVA Allocation Sum = 80% (Below bound) triggers 'Allocations must sum to 100'. (Student Roll No: 24i-3137)

TC ID **TC-BVA-002**

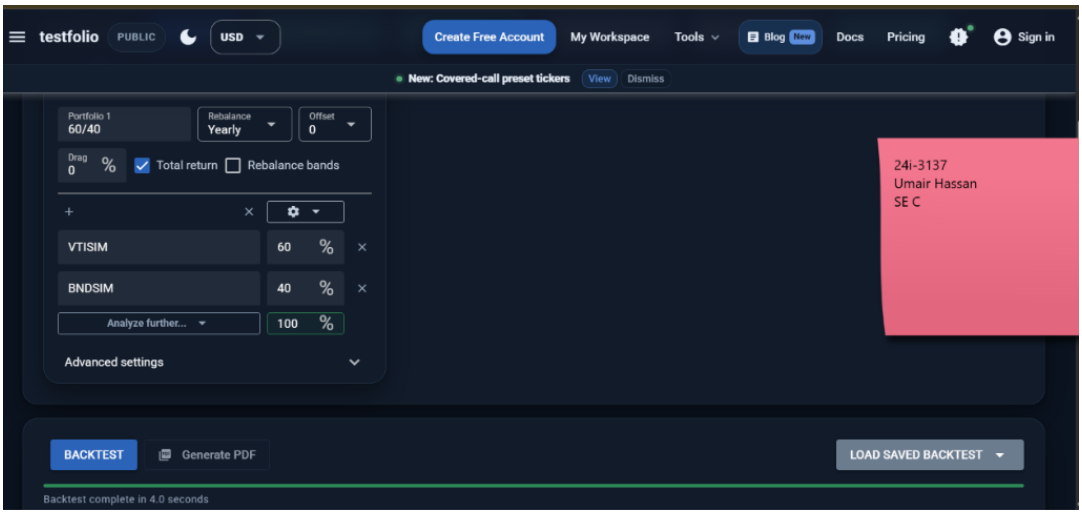


Figure E-45: Portfolio Backtester (/): BVA Allocation Sum = 100% (At bound) executes successfully. (Student Roll No: 24i-3137)

TC ID **TC-BVA-003**

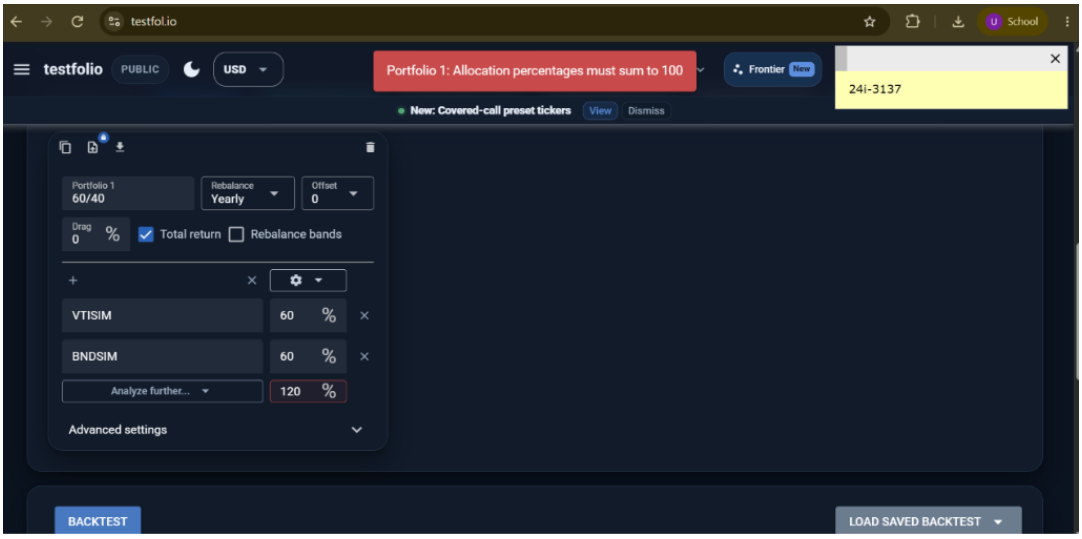


Figure E-46: Portfolio Backtester (/): BVA Allocation Sum = 120% (Above bound) triggers 'Allocations must sum to 100'. (Student Roll No: 24i-3137)

TC ID **TC-BVA-004**

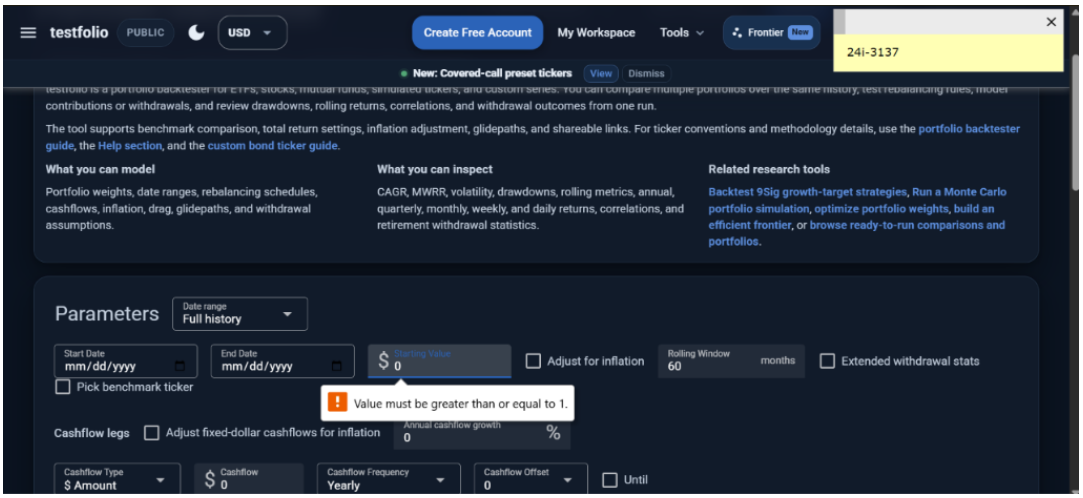


Figure E-47: Portfolio Backtester (/): BVA Starting Value = $0 (Below bound) triggers tooltip 'Value must be greater than or equal to 1.' (Student Roll No: 24i-3137)

TC ID **TC-BVA-006**

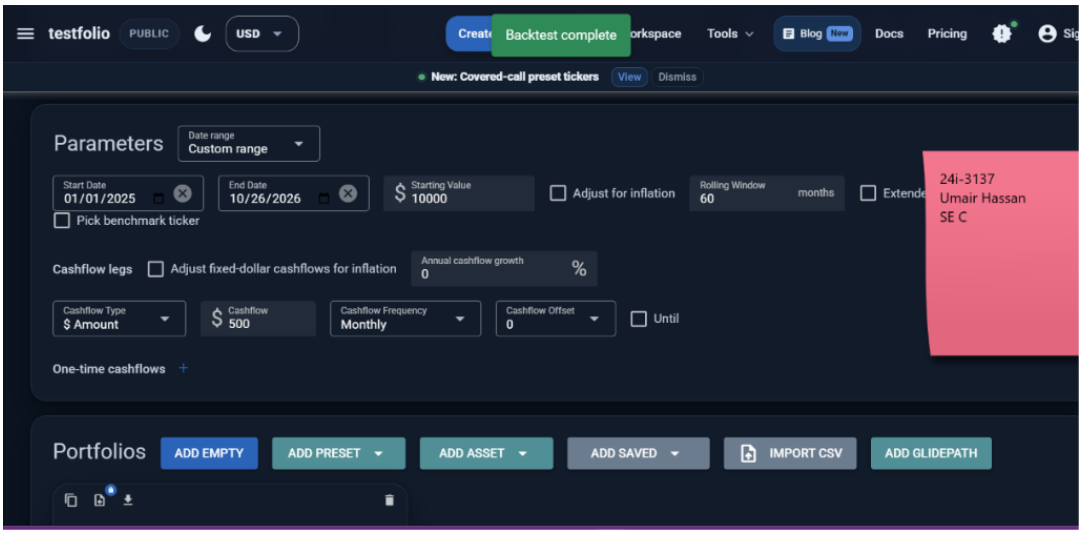


Figure E-48: Portfolio Backtester (/): BVA Starting Value = $10,000 (Above bound) executes successfully. (Student Roll No: 24i-3137)

TC ID **TC-BVA-007**

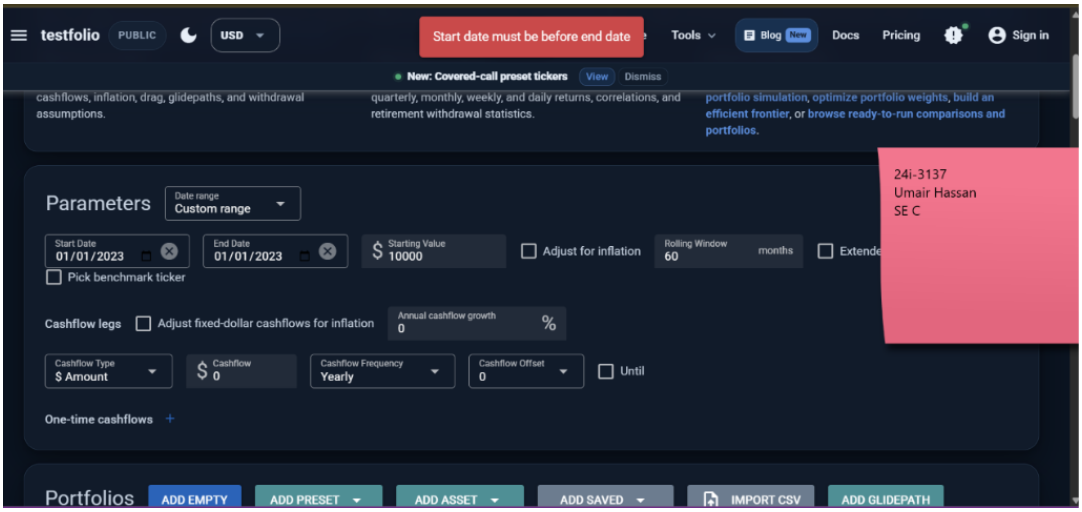


Figure E-49: Portfolio Backtester (/): BVA Dates Start == End (Boundary) triggers 'Start date must be before end date.' (Student Roll No: 24i-3137)

TC ID **TC-BVA-008**

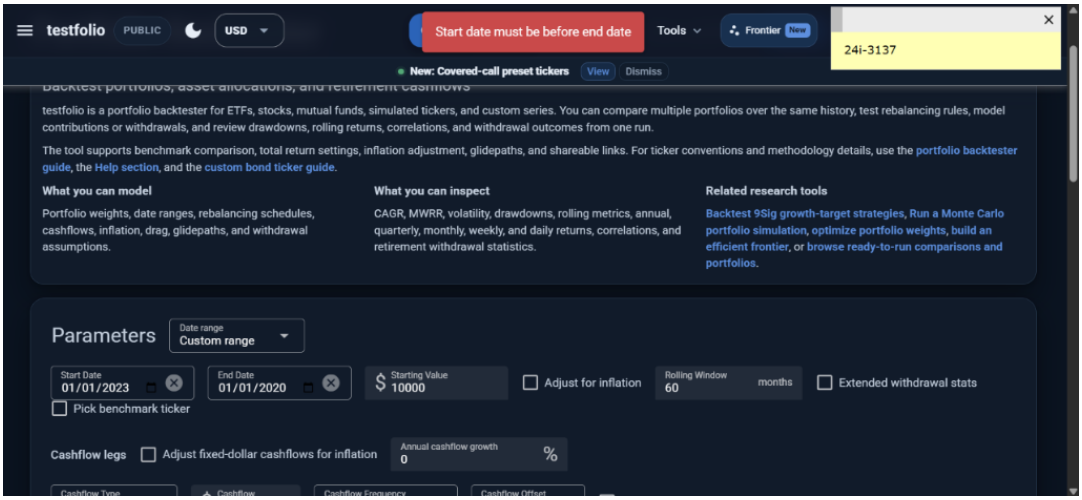


Figure E-50: Portfolio Backtester (/): BVA Dates Start > End (Below bound) triggers 'Start date must be before end date.' (Student Roll No: 24i-3137)

TC ID **TC-BVA-009**

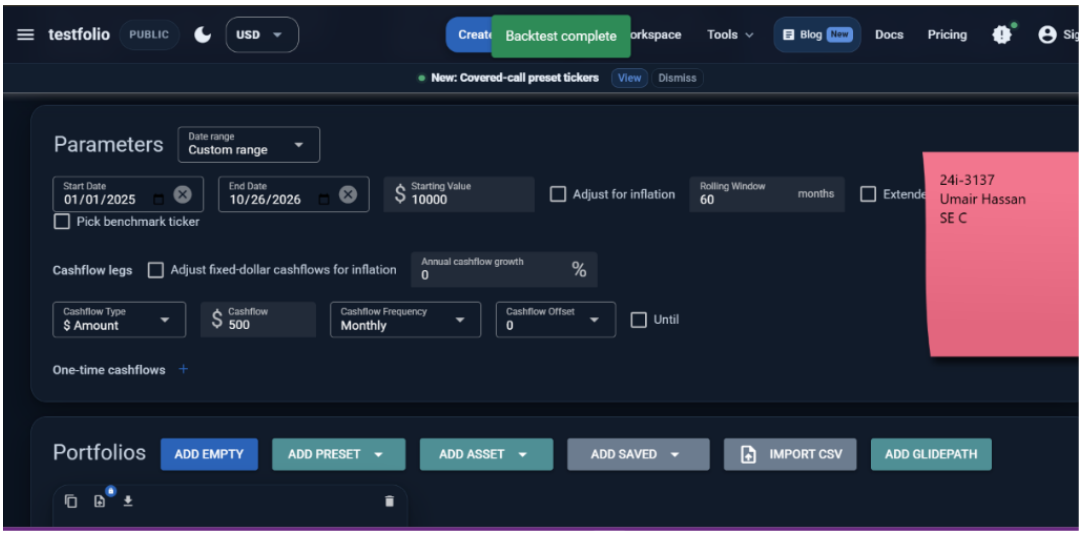


Figure E-51: Portfolio Backtester (/): BVA Dates Start < End (Above bound) executes successfully. (Student Roll No: 24i-3137)

TC ID **TC-BVA-010**

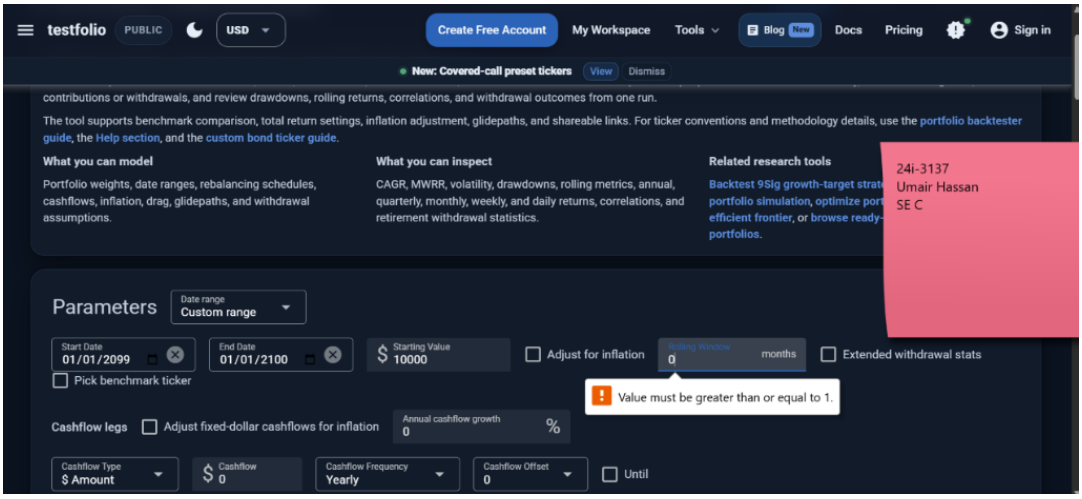


Figure E-52: Portfolio Backtester (/): BVA Rolling Window = 0 (Below bound) triggers 'Value must be greater than or equal to 1.' (Student Roll No: 24i-3137)

TC ID **TC-BVA-012**

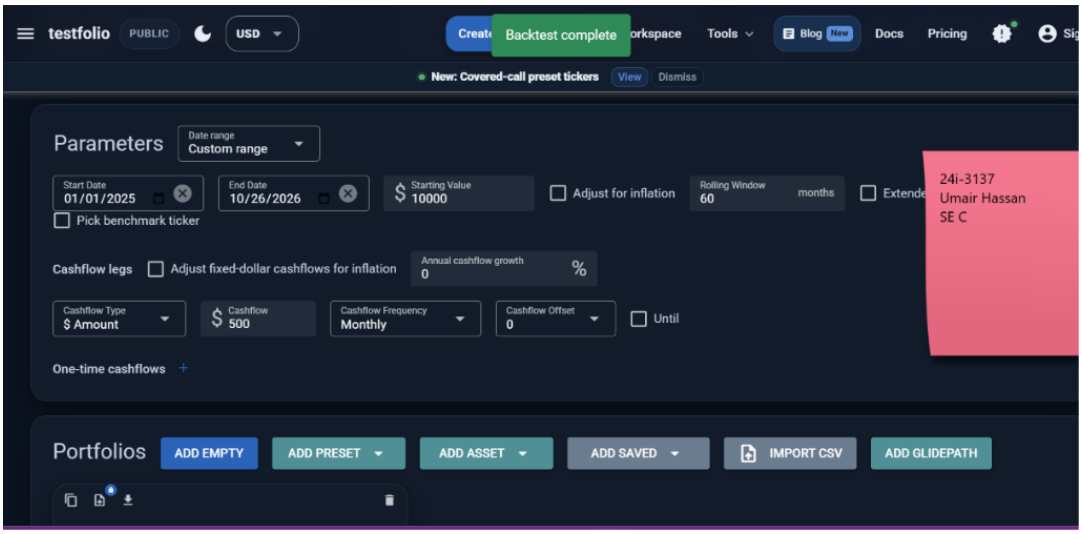


Figure E-53: Portfolio Backtester (/): BVA Rolling Window = 60 months (Above bound) executes successfully. (Student Roll No: 24i-3137)

TC ID **TC-BVA-013**

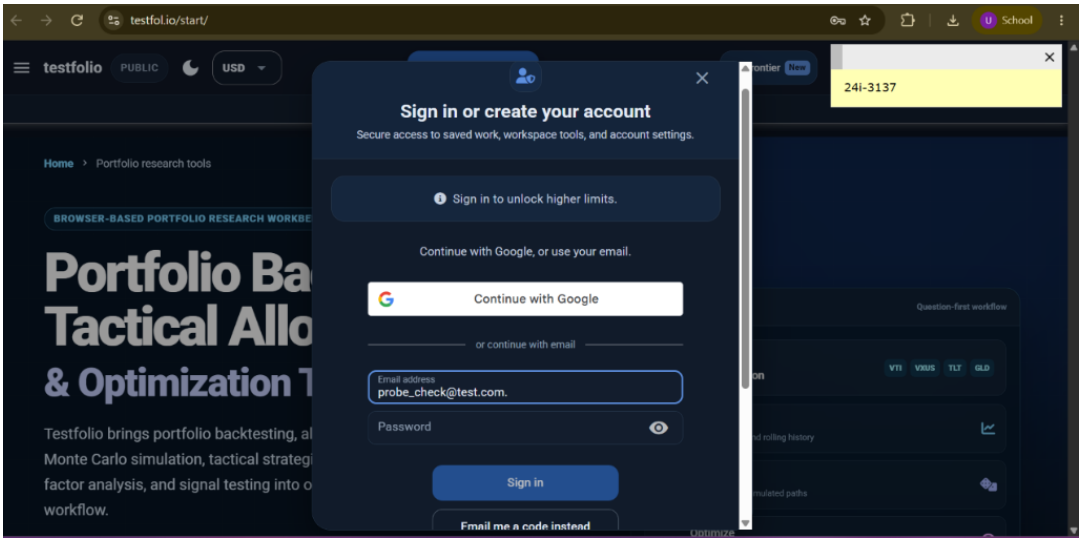


Figure E-54: Auth Modal (/start): BVA Password = 0 chars (Below bound) disables submit button client-side. (Student Roll No: 24i-3137)

TC ID **TC-BVA-014**

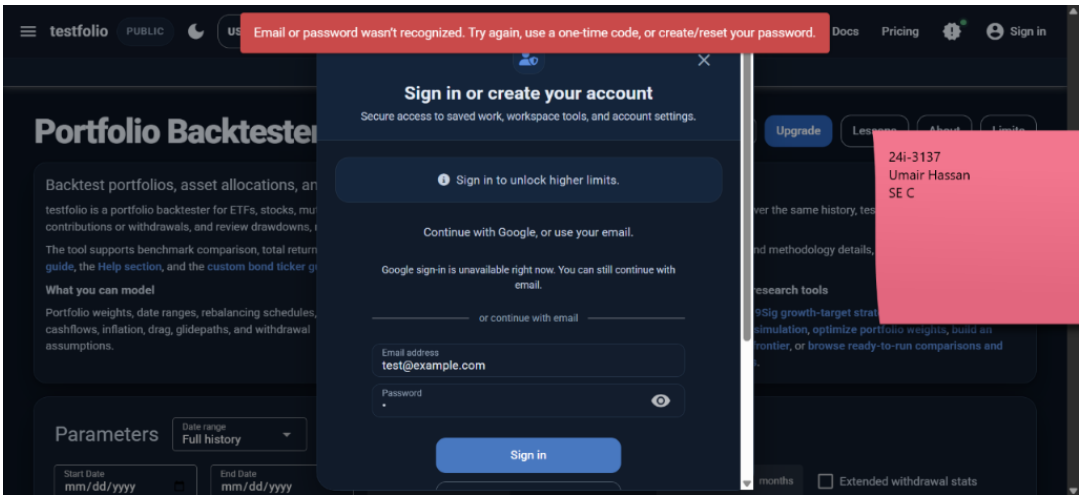


Figure E-55: Auth Modal (/start): BVA Password = 1 char (At bound) enables submit button client-side. (Student Roll No: 24i-3137)

TC ID **TC-BVA-015**

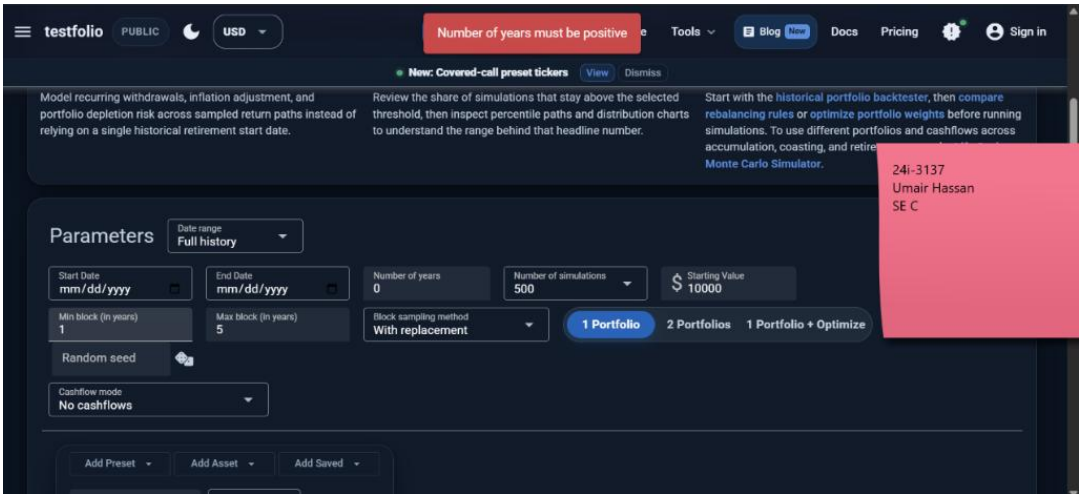


Figure E-56: Monte Carlo Simulator (/monte-carlo): BVA Horizon = 0 yrs (Below bound) triggers 'Number of years must be positive.' (Student Roll No: 24i-3137)

TC ID **TC-BVA-016**

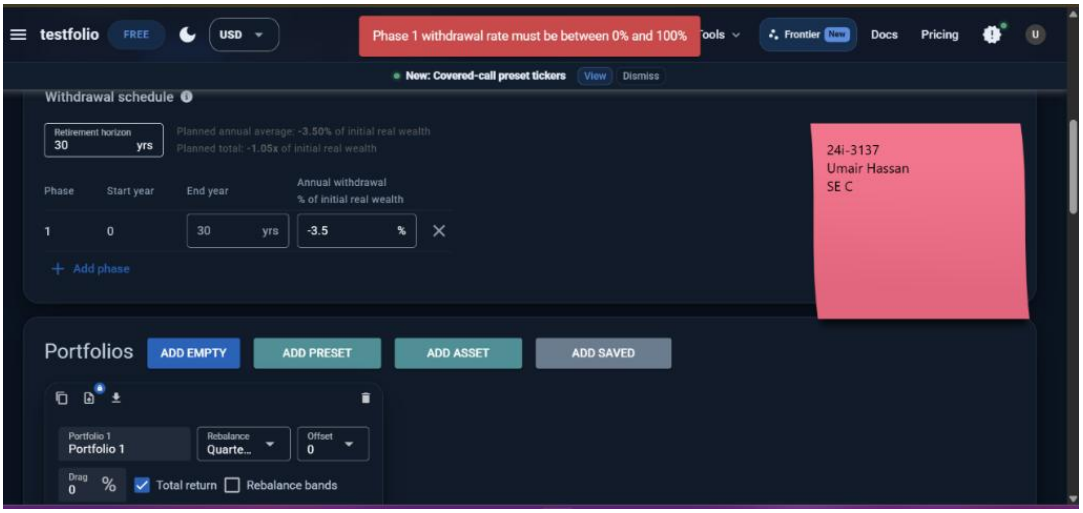


Figure E-57: Safe Withdrawal Rate: BVA Rate = -3.5% (Below bound) triggers 'Phase 1 withdrawal rate must be between 0% and 100%'. (Student Roll No: 24i-3137)

TC ID **TC-BVA-017**

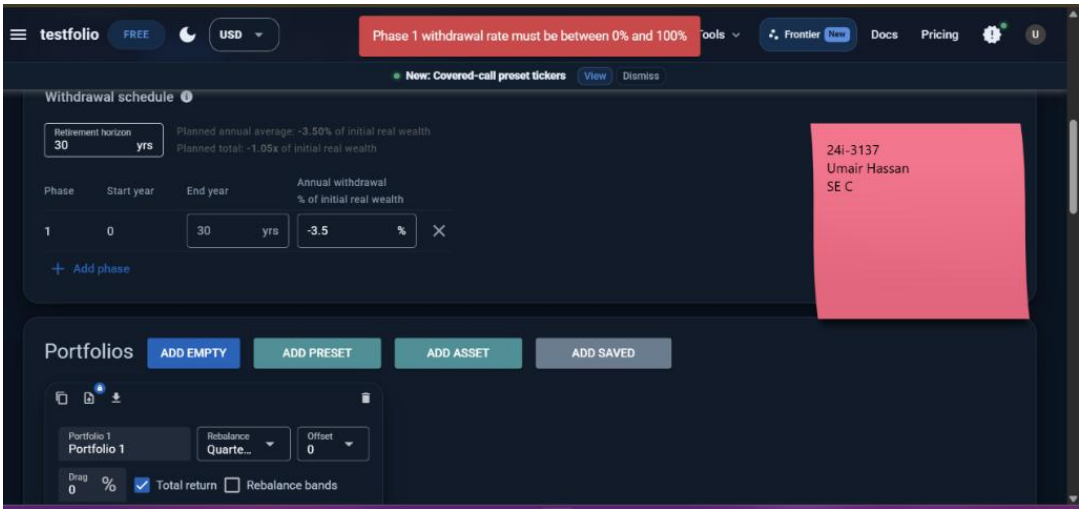


Figure E-58: Safe Withdrawal Rate: BVA Rate = -3.5% (Below bound) triggers 'Phase 1 withdrawal rate must be between 0% and 100%'. (Student Roll No: 24i-3137)

TC ID **TC-BVA-018**

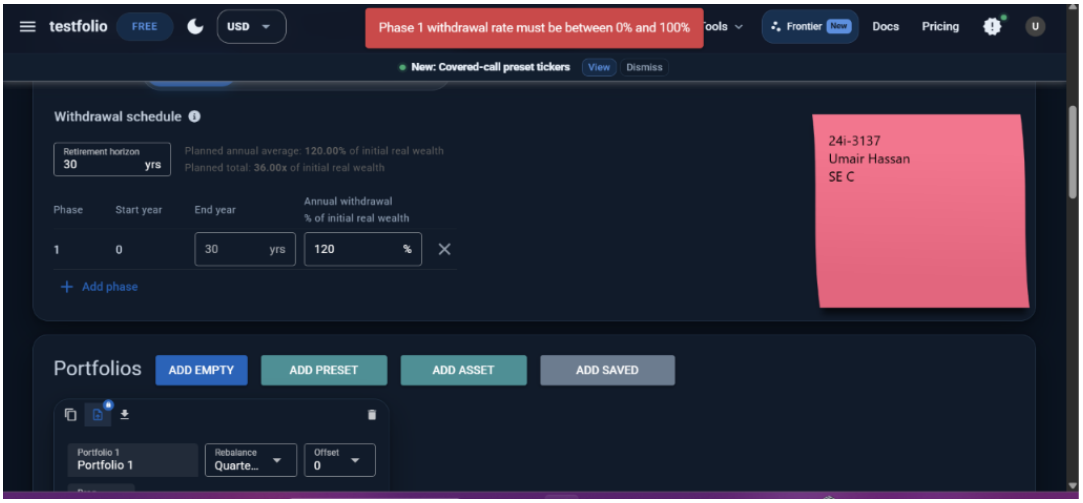


Figure E-59: Safe Withdrawal Rate: BVA Rate = 120% (Above bound) triggers 'Phase 1 withdrawal rate must be between 0% and 100%'. (Student Roll No: 24i-3137)

TC ID **TC-BVA-019**

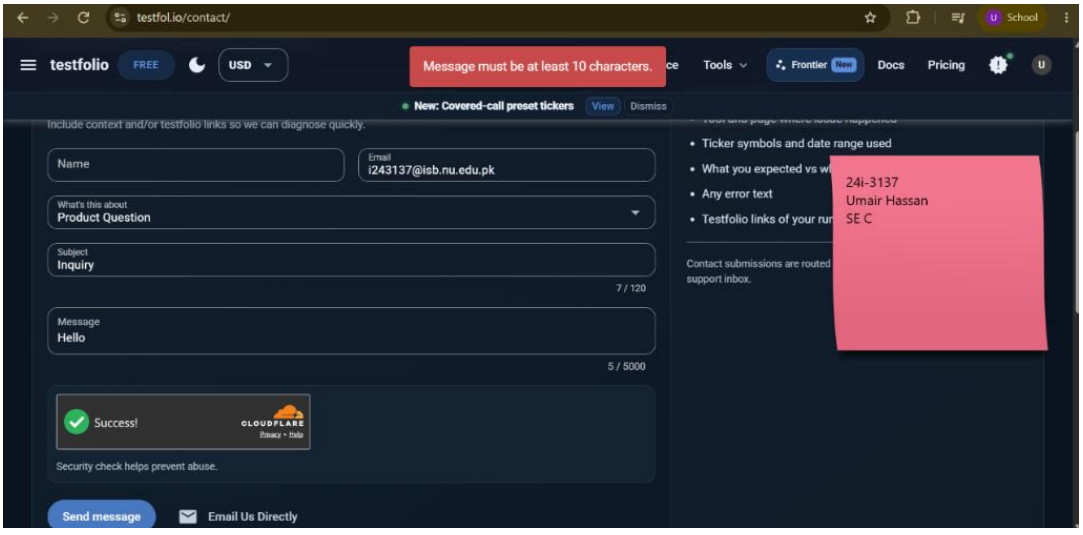


Figure E-60: Contact Form (/contact): BVA Message = 5 chars (Below bound) triggers 'Message must be at least 10 characters.' (Student Roll No: 24i-3137)